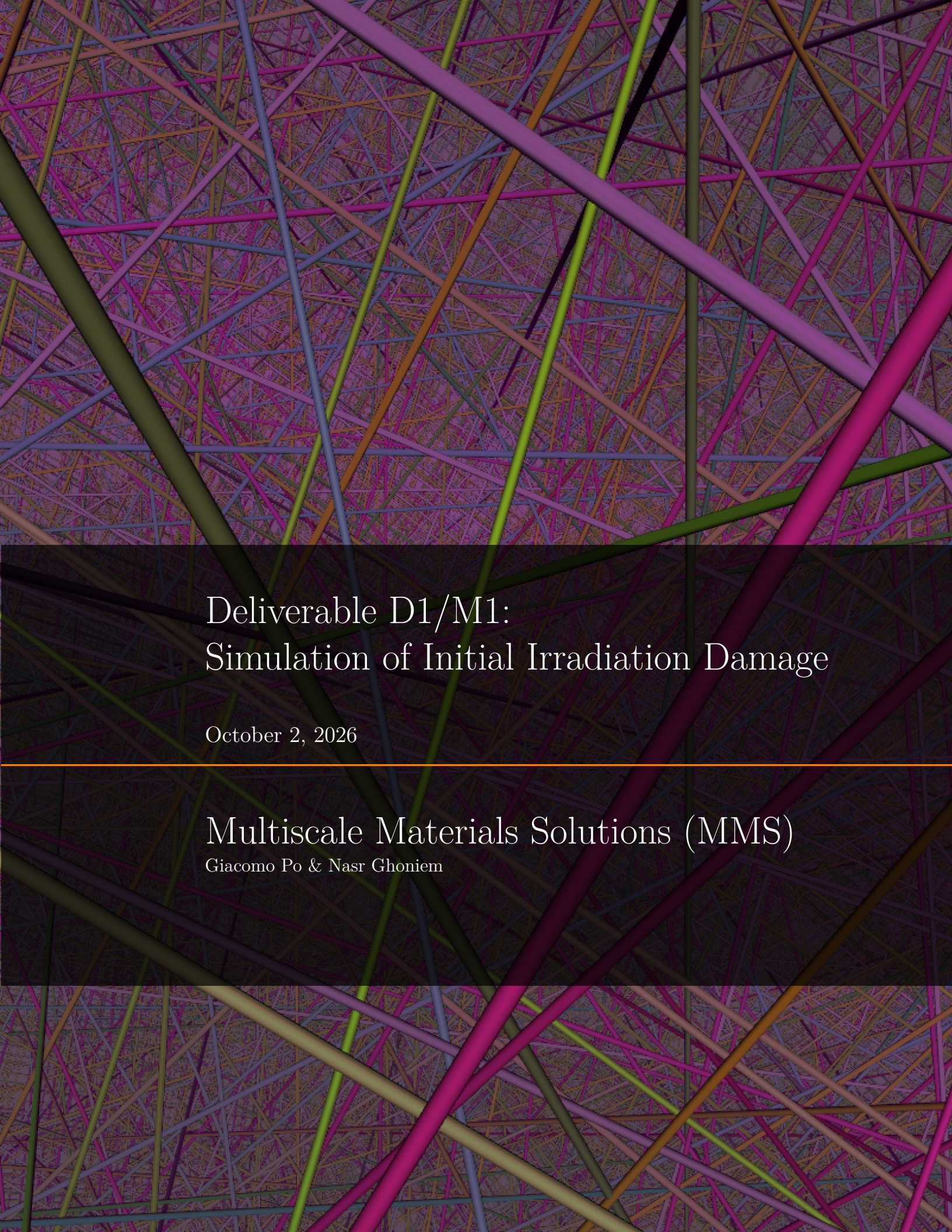


The background of the slide is a dense, abstract pattern of thin, overlapping lines in various colors including purple, blue, green, yellow, and orange. These lines are oriented in different directions, creating a complex, woven texture. Overlaid on this background are several thick, solid-colored lines in shades of purple, blue, green, and yellow, which are oriented diagonally and intersect with the thinner lines.

Deliverable D1/M1: Simulation of Initial Irradiation Damage

October 2, 2026

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Abstract

This report documents progress on Deliverable D1, Multiscale Simulation Methodology, and on Milestone D1/M1, Simulation of Initial Irradiation Damage, whose objective is a spatially resolved Cluster Dynamics–Discrete Dislocation Dynamics framework for irradiation-induced defects in zirconium. Two independent software repositories support this work. **ZrMicro** provides a spatially uniform (0D) cluster-dynamics model and tools for identifying its parameters from measured loop densities and diameters. **MoDELib2** performs three-dimensional spatially resolved simulations of mobile defects and immobile loop populations. Twenty-three parameters are fitted using the inexpensive 0D model, and the calibrated values are transferred to **MoDELib2** through the material file `Zr_CD4opt.txt`. The two repositories, the calibrated material file, and the simulation results constitute the deliverables. The present calibration assumes isotropic diffusion and will be updated for anisotropic diffusion in the next deliverable.

The calibrated reduced model reproduces the build-up, overshoot and saturation of the defect kinetics, the prismatic $\langle a \rangle$ densities and their thermal suppression, and the saturated $\langle a \rangle$ diameter at low to intermediate temperature. The principal quantitative gap is the basal $\langle c \rangle$ diameter, which saturates below the measured value although its number density is matched, pointing to missing physics in vacancy-loop coarsening. The transfer of that calibration is demonstrated rather than assumed: **MoDELib2** reproduces the **ZrMicro** rate coefficients to within 0.1% and the interior mobile concentrations and $\langle c \rangle$ density to within a few percent, every verification ratio staying constant in dose from about 5 dpa — the signature of a structural difference rather than a unit or rate-constant error, which the loops would integrate into a drift. The spatial predictions, interpreted only then, resolve what the mean field averages away: a denuded zone at the absorbing grain boundary that thickens with dose, many small loops near the boundary against fewer large ones in the interior, and stored content correspondingly enhanced at the boundary for the prismatic families and depleted there for the basal one, which carries into grain-size-dependent irradiation growth. The residual code-to-code differences are diagnosed rather than resolved: the $\langle a \rangle$ density in **MoDELib2** sits at about half the reduced-model value, consistent in sign with a homogeneous clustering flux not yet wired into loop nucleation, and, grain boundaries not yet being loop sinks, the near-boundary $\langle a \rangle$ density rises without bound and is not quantitative. The calculations are further restricted applied stress, isotropic diffusion and a bias fitted at 573 K, so the tensorial anisotropy mechanism is not yet exercised. Cross-verification itself proved diagnostic, exposing a Burgers-vector error and an uninitialized binding-energy read upstream.

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1 Introduction

The first deliverable under the current SOW is Deliverable D1: Multiscale Simulation Methodology. The objective is to develop and/or extend a Spatially Resolved Cluster Dynamics (SRCD)–Discrete Dislocation Dynamics (DDD) framework to simulate the nucleation and growth of irradiation-induced defect species in Zirconium. The framework shall generate continuous spatial fields describing defect species concentrations as functions of temperature and irradiation dose. Model parameters shall be calibrated to capture representative defect species and behaviors observed in Zirconium and its alloys. The methodology combines CD simulations to establish the irradiation-induced initial microstructure with a nucleation criterion for $\langle a \rangle$ dislocation loops once a threshold damage level (dpa) is reached.

In milestone D1/M1: Simulation of Initial Irradiation Damage, spatially resolved CD simulations shall be used to predict microstructural evolution, including defect nucleation and growth. Outputs shall consist of continuous spatial fields of defect cluster concentrations at prescribed temperatures and irradiation doses. Results shall be validated against experimental data across a range of irradiation conditions. We report here on our progress in the implementation of deliverable D1, specifically milestone 1 (D1/M1). The roles of the two repositories and the calibration workflow are summarized in Section 2.1.

2 Spatially Resolved Cluster Dynamics

MoDELlib2 implements a SRCD model of the irradiation defect microstructure, where defect clusters are represented by continuum fields. The method is called “**reduced**” because it resolves the concentration of a limited number of **mobile** defect clusters, each having a fixed defect content. Clusters of larger sizes, which are considered to be **immobile**, have their spatial description limited to their number density and total defect content, without resolving their size distribution explicitly.

2.1 Software repositories and calibration workflow

The software delivered with this report consists of two independent repositories, **ZrMicro** and **MoDELlib2**, with complementary roles. **ZrMicro** implements a spatially uniform (0D) cluster-dynamics model and provides the tools for identifying its parameters from experimental loop densities and sizes. The low computational cost of the 0D model makes the repeated simulations required for parameter identification practical.

The calibrated parameters are written to the material file **Zr_CD4opt.txt**, which supplies the material inputs for **MoDELlib2**. This separate repository implements the spatially resolved cluster-dynamics equations and uses the calibrated file to simulate defect nucleation and growth within finite grains, including the effects of absorbing grain boundaries. The workflow therefore proceeds from parameter identification in **ZrMicro** to spatially resolved simulation in **MoDELlib2**. The present calibration is preliminary and assumes isotropic diffusion. It will be updated in the next deliverable to account for anisotropic diffusion.

The deliverables accompanying this report are the two software repositories, the calibrated material file **Zr_CD4opt.txt**, and the simulation results presented here. NNL can use them together by updating its **MoDELlib2** repository, obtaining the independent **ZrMicro** repository, and using the calibrated material file for the spatial simulations. The remainder of this section describes the spatially resolved model; the 0D model and its parameter identification procedure are detailed in the following sections.

2.2 Mobile defect clusters

The mobile defect clusters are defined by the **mobileSpeciesVector**, a vector parameter set in the material file. For example, **Zr_CD4opt.txt** defines **mobileSpeciesVector=-1 1 2 3**; so that four mobile species are considered: single vacancies, self-interstitials, di-interstitials, and tri-interstitials. We denote this vector $\mathbf{m}_M$; its entries are the signed cluster sizes m^k , with the polarity $p^k = m^k/|m^k| \in \{+1, -1\}$ selecting interstitial ($p = +1$) or vacancy ($p = -1$) character, and $|m^k|$ the number of point defects in the cluster. For the example above $\mathbf{m}_M = [-1, +1, +2, +3]^T$ and $\mathbf{p}_M = [-1, +1, +1, +1]^T$. As in Section 2.4, we use the boolean selectors $\mathcal{M}_i = \frac{1}{2}(1 + \mathbf{p}_M)$ and $\mathcal{M}_v = \frac{1}{2}(1 - \mathbf{p}_M)$.

MoDELib2 solves the following steady-state reaction-diffusion partial differential equation (PDE) for the mobile content field $\mathbf{c}_M = [c^v, c^i, c^{2i}, c^{3i}]^T$, expressed in fractional (per-atom) units:

$$\mathbf{0} = -\nabla \cdot \mathbf{J} + \mathbf{P}_M + \frac{1}{2} (\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M + (\mathbf{E} - \mathbf{D}) \mathbf{c}_M + \mathbf{S}_{vL}. \quad (1)$$

Throughout this work, we distinguish two kinds of non-scalar quantities by their typeface. Upright sans-serif bold denotes arrays in species space (indexed by the mobile defect species $v, i, 2i, 3i$ or by loop family), with lowercase symbols for vectors, e.g. the concentration array $\mathbf{c}_M$, the stacked flux $\mathbf{J}$, and the vacancy-loop dissolution source $\mathbf{S}_{vL}$ (Eq. 55), and uppercase symbols for matrices, e.g. the reaction matrix $\mathbf{D}$, the emission matrix $\mathbf{E}$, and the second-order reaction matrix $\mathbf{R}_2$. Bold-italic (e.g. $\mathbf{J}^x$, $\mathbf{D}^x$, ∇ , σ) is reserved for ordinary vectors and tensors in Cartesian space. Thus, the entries $\mathbf{J}^x$ of the species-space flux array $\mathbf{J}$ are themselves Cartesian vectors, and the divergence $\nabla \cdot$ acts on each entry.

The stacked species flux collects the four Cartesian fluxes of the mobile species,

$$\mathbf{J} = \begin{bmatrix} \mathbf{J}^v \\ \mathbf{J}^i \\ \mathbf{J}^{2i} \\ \mathbf{J}^{3i} \end{bmatrix}. \quad (2)$$

Here $\mathbf{J}^x = -\mathbf{D}^x \nabla c^x$ is the diffusive flux of species x , with a possibly-anisotropic diffusion tensor $D_{ij}^x = D_{0ij}^x \exp(-E_{ij}^{xm}/k_B T)$ whose pre-exponential factors D_{0ij}^x and migration energies E_{ij}^{xm} are the vectors of the material matrix `mobileSpeciesD0_SI` and `mobileSpeciesEnergyMigration_eV`.

The vector $\mathbf{P}_M$ is the cascade production term,

$$\mathbf{P}_M = G \mathbf{f}, \quad (3)$$

where G is the nominal dpa rate (`doseRate_dpaPerSec`) and $\mathbf{f}$ is the fraction vector partitioning cascade production among the mobile clusters¹.

The $\mathbf{R}_2$ -term represents second-order reactions between mobile species, modeling the collision of two mobile species to form a third species. The second-order reaction matrix, whose rows contract the concentration array with the per-species reaction matrices $\mathbf{R}_2^x$ of Eq. (6), is

$$\mathbf{R}_2 = \begin{bmatrix} \mathbf{R}_2^v \\ \mathbf{R}_2^i \\ \mathbf{R}_2^{2i} \\ \mathbf{R}_2^{3i} \end{bmatrix}. \quad (4)$$

In Eq. (1) this stacked array enters through the contraction $\frac{1}{2} (\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M$, where, in the case of four mobile species, $\mathbf{I}$ is the identity 4×4 and $\otimes$ the Kronecker product:

$$\mathbf{I} \otimes \mathbf{c}_M^T = \begin{bmatrix} \mathbf{c}_M^T & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{c}_M^T & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{c}_M^T & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{c}_M^T \end{bmatrix}. \quad (5)$$

Note that the x -th row in (1) corresponding to the second-order reaction term is $\frac{1}{2} \mathbf{c}_M^T \mathbf{R}_2^x \mathbf{c}_M$, so this term is a vector in species space. For the chosen vector of mobile species, we have

$$\begin{aligned} \mathbf{R}_2^v &= - \begin{bmatrix} 0 & \beta^{v,i} & \beta^{v,2i} & \beta^{v,3i} \\ \beta^{v,i} & 0 & 0 & 0 \\ \beta^{v,2i} & 0 & 0 & 0 \\ \beta^{v,3i} & 0 & 0 & 0 \end{bmatrix}, & \mathbf{R}_2^i &= - \begin{bmatrix} 0 & \beta^{v,i} & -\beta^{v,2i} & 0 \\ \beta^{v,i} & 2\beta^{i,i} & \beta^{i,2i} & \beta^{i,3i} \\ -\beta^{v,2i} & \beta^{i,2i} & 0 & 0 \\ 0 & \beta^{i,3i} & 0 & 0 \end{bmatrix}, \\ \mathbf{R}_2^{2i} &= -2 \begin{bmatrix} 0 & 0 & \beta^{v,2i} & -\beta^{v,3i} \\ 0 & -\beta^{i,i} & \beta^{i,2i} & 0 \\ \beta^{v,2i} & \beta^{i,2i} & 2\beta^{2i,2i} & \beta^{2i,3i} \\ -\beta^{v,3i} & 0 & \beta^{2i,3i} & 0 \end{bmatrix}, & \mathbf{R}_2^{3i} &= -3 \begin{bmatrix} 0 & 0 & 0 & \beta^{v,3i} \\ 0 & 0 & -\beta^{i,2i} & \beta^{i,3i} \\ 0 & -\beta^{i,2i} & 0 & \beta^{2i,3i} \\ \beta^{v,3i} & \beta^{i,3i} & \beta^{2i,3i} & 2\beta^{3i,3i} \end{bmatrix}, \end{aligned} \quad (6)$$

¹Expanded component-wise, and accounting for the per-cluster entry of the di- and tri-interstitial sources ($G_{2i}/2$ and $G_{3i}/3$), $\mathbf{P}_M$ reproduces the 0D cascade sources $\{G_v, G_i, \frac{1}{2}G_{2i}, \frac{1}{3}G_{3i}\}$ of Section 3.1, with $G_x = G\varepsilon_x$ the corresponding partition of the displacement rate.

where $\beta^{m,n}$ in Eq. (6) represents the reaction rates between species m and n . A positive reaction rate indicates production, while a negative sign indicates consumption. Each rate constant is built from the encounter geometry of the two reactants,

$$\beta^{m,n} = \frac{A^{m,n}}{\Omega_0 |\mathbf{m}_{Mm} \mathbf{m}_{Mn}|} (\bar{D}^m + \bar{D}^n) (r^m + r^n), \quad (7)$$

that is, the sum of the two reactant orientation averaged diffusivities² weighted by their combined capture radius $r^m + r^n$, with $A^{m,n}$ a geometric prefactor defined by the input variable `reactionPrefactorMap`, and Ω_0 the atomic volume. The capture radius r^m used to build $\beta^{m,n}$ follows the point-defect / prismatic-loop geometry

$$r^m = \begin{cases} \left(\frac{3\Omega_0}{4\pi} \right)^{1/3}, & |\mathbf{m}_{Mm}| = 1, \\ \left(\frac{|\mathbf{m}_{Mm}| \Omega_0}{\pi b} \right)^{1/2}, & |\mathbf{m}_{Mm}| > 1. \end{cases} \quad (10)$$

The **E**-term represents the thermal dissociation (emission) of mobile species³. It is the matrix

$$\mathbf{E} = - \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\alpha^{i,2i} & -\frac{1}{3}\alpha^{i,3i} \\ 0 & 0 & \alpha^{i,2i} & -\frac{2}{3}\alpha^{i,3i} \\ 0 & 0 & 0 & \alpha^{i,3i} \end{bmatrix}, \quad (11)$$

where $\alpha^{x,nx}$ is the dissociation rate of an nx -cluster emitting a point defect of type x ,

$$\alpha^{m,n} = \beta^{m,n} \exp \left(-\frac{E_{m,n}^b}{k_B T} \right), \quad (12)$$

With $E_{m,n}^b$, the binding energy of the $(m+n)$ -size cluster (`mobileSpeciesBindingEnergy_eV`).

The **D**-term is the linear reaction matrix that governs the annihilation of mobile defects at immobile sinks. It absorbs the mobile diffusivities into the sink strengths, which themselves combine the network sinks with the mass-conserving loop absorption. The mobile-to-immobile reaction matrix is:

$$\mathbf{D} = \text{diag}\{\boldsymbol{\omega} \odot \mathbf{C}_s\} + \mathbf{D}_d \mathbf{K}_{\mathcal{L}}, \quad \mathbf{D}_d = \text{diag}\{\bar{D}^v, \bar{D}^i, \bar{D}^{2i}, \bar{D}^{3i}\} \quad (13)$$

where $\mathbf{C}_s = [C_s^v, C_s^i, C_s^i, C_s^i]^T$ are the network-only sink concentrations of Section 3.1 ($C_s^v = \frac{a^2}{z_c} \rho_N$, $C_s^i = \frac{a^2}{z_c} Z_N \rho_N$) and $\boldsymbol{\omega} = [\omega_v, \omega_i, \omega_i, \omega_i]^T$. The first term reproduces the network sink reactions $R_{x,s} = \omega_x C_x C_s^x$, while $\mathbf{K}_{\mathcal{L}}$ carries the loop-absorption debits `loop_abs_i`, `loop_abs_v` defined in Eq. (19), so that mobile losses to loops match exactly the content gained by the immobile families and no defect is double-counted. The network term is already a reaction rate, whereas the loop sink strengths

²

The orientation-averaged diffusivity is $\bar{D}^x = \sqrt[3]{\det(\mathbf{D}^x)}$. This quantity is related to the scalar jump frequency ω^m by the relation

$$\bar{D}^x = \frac{a^2}{z_c} \omega_x \quad (8)$$

where a is the lattice constant and z_c the combinatorial factor of the jump. The jump frequencies enter below in three places: the pairwise reaction rates $\beta^{m,n}$ of Eq. (9), the network sink term of Eq. (13), and the point-defect fluxes ϕ_v and ϕ_i that drive loop growth in Eq. (42). Using Eq. (8) makes $\beta^{m,n}$ proportional to the sum of the two reactant jump frequencies. In the material file calibrated from the 0D model (`Zr_CD4opt.txt`) the prefactors $A^{m,n}$ are chosen, by matching the jump-frequency and diffusivity-based reaction coefficients, so that the constant of proportionality is unity:

$$\beta^{m,n} = \omega_m + \omega_n, \quad (9)$$

since the form that the 0D model of Section 3.1 uses for its pairwise reaction rates is $R_{a,b} = (\omega_a + \omega_b) C_a C_b$, and the two constructions agree to within 0.1% on the assembled coefficients (Table 2). Two channels carry extra factors: the recombination-volume factor β_r on the interstitial-vacancy channel, $\beta^{v,i} \rightarrow \beta_r(\omega_i + \omega_v)$, and the combinatorial factors retained on the like-species reactions, $R_{i,i} = 2\omega_i C_i^2$ and $R_{2i,2i} = 4\omega_{2i} C_{2i}^2$. Expanding $\frac{1}{2} \mathbf{c}_M^T \mathbf{R}_2^x \mathbf{c}_M$ component-wise then reproduces the 0D pair-reaction budget of each monomer and small-cluster equation; the interstitial line, for instance, yields $R_{2i,v} + 2\mathcal{E}_{2i} + 3\mathcal{E}_{3i} - (R_{i,v} + R_{i,i} + R_{i,2i} + R_{i,3i})$ once the emission and sink terms below are added.

³Component-wise, $\mathbf{E} \mathbf{c}_M$ reproduces the 0D thermal-emission terms $\mathcal{E}_{2i} = \omega_{2i} e_{2i} C_{2i}$ and $\mathcal{E}_{3i} = \omega_{3i} e_{3i} C_{3i}$ of Section 3.1.

acquire the mobile diffusivities, so both contributions have units of frequency (Hz), and $\overline{D}^m$ is the (orientationally averaged) diffusion coefficient of mobile species m .

Here $\odot$ denotes the Hadamard (component-wise) product of two species-space vectors, so that the first term of Eq. (13) is the diagonal matrix

$$\text{diag}\{\boldsymbol{\omega} \odot \mathbf{C}_s\} = \begin{bmatrix} \omega_v C_s^v & 0 & 0 & 0 \\ 0 & \omega_i C_s^i & 0 & 0 \\ 0 & 0 & \omega_i C_s^i & 0 \\ 0 & 0 & 0 & \omega_i C_s^i \end{bmatrix}. \quad (14)$$

Its x -th diagonal entry multiplies c^x in Eq. (1) and so returns the network sink reaction $R_{x,s} = \omega_x c^x C_s^x$ species by species. The three interstitial species share the monomer jump frequency ω_i and the interstitial sink concentration C_s^i , which is why the last three diagonal entries are equal.

Finally, the term $\mathbf{S}_{vL}$ represents a source of mobile defects originating from the annealing of immobile defects, and it will be defined later in Eq. (55).

The mobile species PDE (1) is accompanied by the concentration boundary condition⁴ (BC)

$$\mathbf{c}_M(\mathbf{x}) = \mathbf{c}_{\text{bnd,eq}}(\mathbf{x}) \quad (\text{mobile species BC}), \quad (16)$$

where $c_{\text{bnd,eq}}^k(\mathbf{x})$ is the local concentration of the species in equilibrium with the boundary,

$$c_{\text{bnd,eq}}^k(\mathbf{x}) = c_0^k(\mathbf{x}) \exp\left(-\frac{\mathbf{m}_{Mk} t_n(\mathbf{x}) \Omega}{k_B T}\right), \quad (17)$$

in which t_n is the normal traction on the boundary and $c_0^k(\mathbf{x})$ is the thermal-equilibrium concentration

$$c_0^k(\mathbf{x}) = |\mathbf{m}_{Mk}| \exp\left(-\frac{E_k^f + \mathbf{m}_{Mk} q \Omega_k^r}{k_B T}\right), \quad (18)$$

where $q(\mathbf{x}) = -\text{tr}(\boldsymbol{\sigma}(\mathbf{x}))/3$ is the local pressure, $\Omega_k^r = \omega_k^r \Omega$ the relaxation volume of the k -th cluster (with a relative relaxation volume ω_k^r , `mobileSpeciesRelRelaxVol`), and E_k^f its boundary formation energy (`mobileSpeciesEnergyFormation_eV`). The stress terms in Eqs. (17) and (18) are multiplied by the signed cluster size $\mathbf{m}_{Mk}$, rendering opposite effects for interstitial and vacancy-type clusters.

2.3 Bias factors and loop sink strengths

The loop sink-strength block $\mathbf{K}_{\mathcal{L}}$ of Eq. (13) is diagonal in the mobile species and collects, for each mobile defect m , the total sink strength of the immobile populations,

$$\mathbf{K}_{\mathcal{L}} = \text{diag}\{\mathbf{k}^2\}, \quad \mathbf{k}^2 = \frac{2\pi}{\Omega_0} \mathbf{Z}^T (\mathbf{R} \odot \mathbf{n}_I). \quad (19)$$

Here $\mathbf{Z}$ is the matrix of bias factors with components Z_{km} (rows indexed by loop family and columns by mobile species) given in Eq. (21), $\mathbf{R} = [R^k]_{k \in \mathcal{L}}$ collects the effective cluster radii, and $\mathbf{n}_I$ is the cluster density per host atom defined in Section 2.4. The contribution of family k to the sink strength for mobile species m is

$$\langle k_{k,m}^2 \rangle = \frac{2\pi}{\Omega_0} Z_{km} R^k n^k. \quad (20)$$

The total sink strength is $k_m^2 = \sum_{k \in \mathcal{L}} \langle k_{k,m}^2 \rangle$. Since n^k is a number per host atom, n^k/Ω_0 is the number density per unit volume, and each sink-strength contribution has units of inverse length squared. The geometric factor is fixed at 2 for all cluster sizes. The continuous transition from a compact bi-pyramid to a planar loop is represented by the effective radius R^k defined in Eq. (29) of Section 2.4, without a separate shape factor.

⁴Eq. (16) is a pure essential (Dirichlet) BC representing a perfect sink. A more general Robin-type BC

$$-\mathbf{J}^k \cdot \mathbf{n} = h^k [c^k(\mathbf{x}) - c_{\text{bnd,eq}}^k(\mathbf{x})], \quad (15)$$

recovers the perfect sink condition as $h^k \rightarrow \infty$ and the no-flux condition as $h^k \rightarrow 0$.

The capture bias Z_{km} of loop family k for mobile species m is the product of a basic bias factor and a diffusional-anisotropy-difference (DAD) factor,

$$Z_{km} = Z_{km}^0 Z_k^{\text{DAD}}(p^m), \quad p^m = \left(\frac{D_c^m}{D_a^m} \right)^{1/6}. \quad (21)$$

The basic bias factor Z_{km}^0 is supplied by the material parameter `discreteDislocationBias` and may depend on both the loop family and mobile species. The dimensionless multiplier Z_k^{DAD} accounts for diffusion anisotropy and depends on the loop habit:

$$Z_k^{\text{DAD}}(p) = \begin{cases} \frac{p + p^{-2}}{2}, & k \in \{a_1L, a_2L, a_3L\} \quad (\text{prismatic } \langle a \rangle \text{ loops}), \\ p, & k = vL \quad (\text{basal } \langle c \rangle \text{ loops}). \end{cases} \quad (22)$$

For anisotropic diffusion, the same mobile species generally has different DAD multipliers at prismatic and basal loops because their habit planes differ. Here D_a^m and D_c^m are the diffusivities in the basal plane and along the c axis, respectively, obtained from the diffusion tensor $\mathbf{D}^m$ used in the mobile-species flux. For isotropic diffusion, $p^m = 1$ and both multipliers are unity, so $Z_{km} = Z_{km}^0$; the basic bias need not be unity.

The same total capture bias Z_{km} enters the loop-growth rates of Eq. (43) and the mobile-defect sink strengths. Substitution into Eq. (20) gives the prismatic contribution $\langle k_{a_jL,m}^2 \rangle = Z_{a_jL,m}^0 (p^m + (p^m)^{-2}) \pi R^{a_jL} n^{a_jL} / \Omega_0$ and the basal contribution $\langle k_{vL,m}^2 \rangle = 2Z_{vL,m}^0 p^m \pi R^{vL} n^{vL} / \Omega_0$. In the isotropic limit both have the form $2\pi Z_{km}^0 R^k n^k / \Omega_0$. For preferential basal-plane diffusion ($p^m < 1$), the basal DAD multiplier is smaller than the prismatic multiplier; the total sink strengths also depend on the basic bias factors, radii, and densities. The vacancy-cluster sink strength varies continuously through the interpolated radius R^{vL} as the clusters flatten onto the basal plane.

2.4 Immobile defect clusters

In the CD implementation of MoDELb2, interstitial clusters larger than m_{max} and vacancy clusters larger than m_{min} are considered immobile and are treated as dislocation loops. In irradiated Zr four immobile loop families are represented, following Section 3.1: one vacancy ($\langle c \rangle$) loop family and three interstitial ($\langle a \rangle$) loop families indexed by the orientation of their Burgers vector along the three prismatic directions a_1, a_2, a_3 ,

$$\mathcal{L} = \{vL, a_1L, a_2L, a_3L\}, \quad (23)$$

Each family k carries a polarity $p^k \in \{+1, -1\}$ ($p = +1$ interstitial, $p = -1$ vacancy),

$$p^{vL} = -1, \quad p^{a_1L} = p^{a_2L} = p^{a_3L} = +1, \quad (24)$$

defined by the input parameter `immobileSpeciesVector`. The interstitial and vacancy families are selected by the boolean vectors

$$\mathcal{I}_i = \frac{1}{2}(1 + p) \quad \text{and} \quad \mathcal{I}_v = \frac{1}{2}(1 - p). \quad (25)$$

Each immobile family is described by two spatially-resolved fields: the cluster density n^k , i.e. the number of loops of family k per host atom, and the defect content c^k , i.e. the number of point defects stored in that family per host atom,

$$\mathbf{n}_I(\mathbf{x}) = [n^{vL}, n^{a_1L}, n^{a_2L}, n^{a_3L}]^T \quad (\text{cluster density, clusters per atom}) \quad (26)$$

$$\mathbf{c}_I(\mathbf{x}) = [c^{vL}, c^{a_1L}, c^{a_2L}, c^{a_3L}]^T \quad (\text{defect content, defects per atom}). \quad (27)$$

Both are dimensionless (fractional) quantities. The mean number of point defects per cluster of family k follows from the ratio of content to density,

$$m^k = \frac{c^k}{n^k}. \quad (28)$$

The shape of a cluster depends on its size m^k : it has been shown that vacancy clusters transition from a compact bi-pyramid geometry to a planar (platelet) loop with increasing cluster size. We therefore include this transition of clusters from volumetric defects to planar loops, so that the effective radius of family k interpolates between the two limiting geometries,

$$R^k(m^k) = [1 - S^k(m^k)] R_{\text{pyr}}^k + S^k(m^k) R_{\text{loop}}^k, \quad (29)$$

where the pyramid and loop radii are

$$R_{\text{pyr}}^k = \left(\frac{m^k \Omega_0}{\sqrt{8}} \right)^{1/3}, \quad R_{\text{loop}}^k = \sqrt{\frac{m^k \Omega_0}{\pi b^k}}. \quad (30)$$

The function S^k is a sigmoidal transition function operating a smooth switch between the two shapes,

$$S^k(m^k) = \frac{1}{1 + \exp \left[-\frac{m^k - m_0^k}{w^k} \right]}, \quad (31)$$

whose midpoint and steepness are set by

$$m_0^k = \frac{m_{\min}^k + m_{\max}^k}{2} - m_s^k, \quad w^k = w_0 (m_{\max}^k - m_{\min}^k), \quad (32)$$

so that clusters with $m^k \lesssim m_{\min}^k$ are predominantly bi-pyramid-like while clusters with $m^k \gtrsim m_{\max}^k$ are predominantly loop-like. The same sigmoidal function interpolates the cluster formation-volume tensor,

$$\Omega^k(m^k) = (1 - S^k(m^k)) \Omega_{\text{pyr}}^k + S^k(m^k) \Omega_{\text{loop}}^k, \quad (33)$$

with the loop and pyramid limits

$$\Omega_{\text{loop}}^k = \mathcal{I}^k \frac{\mathbf{b}^k \otimes \mathbf{b}^k}{\mathbf{b}^k \cdot \mathbf{b}^k} V^k, \quad \Omega_{\text{pyr}}^k = \mathcal{I}^k \frac{\mathbb{I}}{3} V^k, \quad (34)$$

where $V^k = m^k \Omega_0$ is the cluster volume and $\mathcal{I}^k$ the loop selector. The pyramid regime applies to the sub-critical vacancy ($\langle c \rangle$) family, which collapses smoothly onto the planar vacancy loop as it grows.

Both fields obey spatially-resolved ordinary differential equations coupled to the mobile PDE (1): a density equation for $\mathbf{n}_I$ (nucleation and annealing of loops) and a content equation for $\mathbf{c}_I$ (defect storage through absorption). These are developed in turn below.

2.4.1 The cluster density ODE

The cluster density evolves by nucleation and thermal annealing,

$$\frac{d\mathbf{n}_I}{dt} = \dot{\mathbf{N}}^{\text{nuc}} - \mathbf{a}_n - \Delta \mathbf{N}, \quad (35)$$

where $\dot{\mathbf{N}}^{\text{nuc}}$ is the nucleation-rate vector, $\mathbf{a}_n$ the (vacancy-loop) annealing vector, and $\Delta \mathbf{N}$ the coalescence sink. The coalescence sink is diagonal in the loop family and, for each family $k \in \{vL, a_1L, a_2L, a_3L\}$, reproduces the two-channel geometric coalescence of Section 3.1 (a like-loop channel that conserves content and a loop-network channel that is booked as network absorption),

$$\Delta \mathbf{N} = \text{diag} \{ \nu_{LL} \odot \phi_{LL} + \nu_{LN} \odot \phi_{LN} \} \mathbf{n}_I, \quad (36)$$

where $\odot$ denotes the Hadamard (elementwise) product. The vectors collect the family-wise quantities: ϕ_{LL}^k and ϕ_{LN}^k are the Avrami overlap probabilities for like-loop and loop-network coalescence, respectively, while ν_{LL}^k and ν_{LN}^k are the corresponding rates. The Avrami overlap probabilities and rates are controlled by the input variable `captureZoneFactor`. These quantities are defined in the 0D model in Eqs. (58) and (59); n_I^k is the k -th component of the immobile cluster-density vector $\mathbf{n}_I$, expressed as the number of loops per host atom. The nucleation-rate vector is

$$\dot{\mathbf{N}}^{\text{nuc}} = (\mathbf{s}^{\text{nuc}} + \mathbf{G}_{\mathcal{L}}) \odot \mathbf{n}^{\text{nuc}}, \quad (37)$$

where $\oslash$ denotes elementwise (Hadamard) division and $\mathbf{n}^{\text{nuc}} = [n_{vL}^{\text{nuc}}, n_{ia}^{\text{nuc}}, n_{ia}^{\text{nuc}}, n_{ia}^{\text{nuc}}]^T$ contains the number of defects per newly nucleated cluster in each family. The source vectors $\mathbf{s}^{\text{nuc}}$ and $\mathbf{G}_{\mathcal{L}}$, defined in the following subsection (Section 2.4.2), represent defect content supplied by mobile-cluster reactions and cascade production, respectively. Division by the nucleation sizes converts these content-production rates into cluster-number production rates.

The orientation weights are the stress-induced preferential nucleation (SIPN) fractions, which distribute defects among the immobile families through a Boltzmann factor coupling the applied stress $\boldsymbol{\sigma}$ to each family's formation-volume tensor $\boldsymbol{\Omega}^k$,

$$\chi_i^k = \frac{\exp\left(\frac{\boldsymbol{\sigma} : \boldsymbol{\Omega}^k}{k_B T}\right) \mathcal{I}_i^k}{\sum_p \exp\left(\frac{\boldsymbol{\sigma} : \boldsymbol{\Omega}^p}{k_B T}\right) \mathcal{I}_i^p}, \quad \chi_v^k = \frac{\exp\left(\frac{\boldsymbol{\sigma} : \boldsymbol{\Omega}^k}{k_B T}\right) \mathcal{I}_v^k}{\sum_p \exp\left(\frac{\boldsymbol{\sigma} : \boldsymbol{\Omega}^p}{k_B T}\right) \mathcal{I}_v^p}, \quad (38)$$

where $\mathcal{I}_i^k$ and $\mathcal{I}_v^k$ are the interstitial and vacancy selectors, and the sums run over all immobile families p . At zero stress every $\boldsymbol{\sigma} : \boldsymbol{\Omega}^k$ vanishes and the interstitial partition reduces to the common seed $\frac{1}{3}$ among the three prismatic orientations, recovering $\chi_i^{a_k L}(\mathbf{0}) = \frac{1}{3}$. The annealing vector acts only on the vacancy families,

$$\mathbf{a}_n = \left[\frac{n_I^{vL}}{\tau_{vL}}, 0, 0, 0 \right]^T, \quad (39)$$

with the Arrhenius lifetimes $\tau_{vL}(T) = \tau_{vL}^0 \exp(E_a^{vL}/k_B T)$.

2.4.2 The defect content ODE

The defect content of the loops evolves through mass-conserving absorption of mobile defects, cascade deposition, nucleation seeding, coalescence, dissolution, and thermal emission,

$$\frac{d\mathbf{c}_I}{dt} = \dot{\mathbf{G}}^c + \mathbf{G}_{\mathcal{L}} + \mathbf{s}^{\text{nuc}} - \mathbf{A}_c - \mathbf{A}_c^{\text{emit}} - \Delta\mathbf{C}. \quad (40)$$

The coalescence content sink removes stored content only through the loop-network channel (the like-loop channel merges two loops without losing content), so its vector form is

$$\Delta\mathbf{C} = \text{diag}\{\boldsymbol{\nu}_{LN} \oslash \boldsymbol{\phi}_{LN}\} \mathbf{c}_I. \quad (41)$$

Here $\mathbf{c}_I$ is the immobile defect-content vector. Like-loop coalescence changes the number of clusters while retaining their total content; loop-network coalescence transfers content out of the loop population.

The net content-growth vector $\dot{\mathbf{G}}^c$ follows from the same species-resolved absorption rates that enter the mobile sink term. Define

$$\Gamma_{km} = \overline{D}^m \langle k_{k,m}^2 \rangle c_M^m = \frac{2\pi}{\Omega_0} \overline{D}^m Z_{km} R^k n_I^k c_M^m. \quad (42)$$

The nonnegative rate Γ_{km} counts defects of mobile species m absorbed by family k per host atom per unit time. Since c_M^m is already a defect-content concentration, it is not multiplied again by $|m^m|$. For assembly, the net content-growth term can be written as a diagonal matrix acting on the immobile content vector,

$$\dot{\mathbf{G}}^c = \mathbf{H}_I \mathbf{c}_I, \quad \mathbf{H}_I = \text{diag}\{h_k\}_{k \in \mathcal{L}}, \quad (43)$$

where, for a populated family with $c_I^k > 0$,

$$\begin{aligned} h_k &= \frac{2\pi R^k n_I^k}{\Omega_0 c_I^k} \sum_{m \in \mathcal{M}} p^k p^m \overline{D}^m Z_{km} c_M^m \\ &= \frac{2\pi R^k}{\Omega_0 m_I^k} \sum_{m \in \mathcal{M}} p^k p^m \overline{D}^m Z_{km} c_M^m, \quad m_I^k = \frac{c_I^k}{n_I^k}. \end{aligned} \quad (44)$$

Each diagonal entry has units of inverse time and may be positive (growth) or negative (shrinkage). This factorization is exact, since $h_k c_I^k = \sum_m p^k p^m \Gamma_{km}$. The matrix depends on the current mobile concentrations and immobile state through R^k and m_I^k ; it is therefore a state-dependent coefficient matrix, not the full nonlinear Jacobian. Holding these coefficients fixed during an update gives a linear contribution to the assembled content equation. For an absent family ($n_I^k = c_I^k = 0$), absorption is zero and its diagonal entry can be set to zero; nucleation supplies the initial number and content. A state with $c_I^k = 0$ but $n_I^k > 0$ must be handled through the unfactored rates Γ_{km} and the chosen minimum-size treatment, rather than by dividing by zero.

Absorption of like defects increases the stored content; absorption of opposite defects decreases it by recombination. Thus vacancy absorption grows a vacancy loop, while interstitial absorption shrinks it. The mobile loss for species m is $\sum_k \Gamma_{km}$, which equals the loop contribution to $(\mathbf{Dc}_M)_m$ in Eq. (13). This loss counts all absorption events, not the signed net growth of the loops. If a minimum-size gate is applied to a shrinking channel, the same factor must multiply Γ_{km} in both the mobile loss and the immobile growth terms.

The cascade deposition vector $\mathbf{G}_L$ delivers the loop-borne fraction of the collision cascade,

$$\mathbf{G}_L = G(\varepsilon_{iL}\chi_i + \varepsilon_{vL}\chi_v), \quad \varepsilon_{iL} = 1 - \mathbf{f} \cdot \mathcal{M}_i, \quad \varepsilon_{vL} = 1 - \mathbf{f} \cdot \mathcal{M}_v. \quad (45)$$

The nucleation-seeding vector transfers the defect content lost from the mobile species by clustering into the immobile families,

$$\mathbf{s}^{\text{nuc}} = R_{2,i}\chi_i + R_{2,v}\chi_v. \quad (46)$$

Here $R_{2,i}$ and $R_{2,v}$ are the interstitial and vacancy content-transfer rates, respectively, expressed as defects per host atom per unit time. The orientation weights χ_i and χ_v distribute these rates among families of the corresponding polarity according to Eq. (38).

To compute the transfer rates, first combine the per-species second-order reaction matrices of Eq. (6) into the total signed-content matrix,

$$\mathbf{R}_2^{\text{tot}} = \sum_{k \in \mathcal{M}} p^k \mathbf{R}_2^k, \quad p^k = \frac{m^k}{|m^k|}, \quad (47)$$

where $\mathcal{M}$ is the set of mobile species. The polarity factor counts interstitial content positively and vacancy content negatively. The superscript tot distinguishes this square matrix from the stacked array $\mathbf{R}_2$ in Eq. (4). Since the mobile concentrations are already expressed in defect-content units, no additional cluster-size weighting is required in this sum.

For a pair of mobile reactants p, q , the signed size of the reaction product is $m^p + m^q$. Products above m_{max} enter the immobile interstitial population; products below m_{min} enter the immobile vacancy population. Restricting the quadratic reaction sum to these two sets gives

$$R_{2,i} = -\frac{1}{2} \sum_{m^p + m^q > m_{\text{max}}} (\mathbf{R}_2^{\text{tot}})_{pq} c_M^p c_M^q, \quad (48)$$

and

$$R_{2,v} = \frac{1}{2} \sum_{m^p + m^q < m_{\text{min}}} (\mathbf{R}_2^{\text{tot}})_{pq} c_M^p c_M^q. \quad (49)$$

The factor 1/2 is the same symmetry factor used in the mobile reaction term. The opposite signs convert the loss of positive interstitial content and the loss of negative vacancy content into positive transfer rates. Reactions whose products remain in the mobile size range do not contribute to either immobile source. For the present mobile species $\{-1, 1, 2, 3\}$, $m_{\text{min}} = -1$ and $m_{\text{max}} = 3$; the interstitial sum selects products larger than three defects, while a vacancy transfer requires a vacancy-clustering reaction included in the reaction matrices.

The content-annealing vector $\mathbf{A}_c$ is determined by the same cluster-removal process as $\mathbf{a}_n$ in Eq. (39). Within the two-moment description, each disappearing cluster removes its family's mean defect content $m^k = c_I^k/n_I^k$, so

$$\mathbf{A}_c = \mathbf{m}_I \odot \mathbf{a}_n, \quad \mathbf{m}_I = \mathbf{c}_I \oslash \mathbf{n}_I. \quad (50)$$

Here $\odot$ and $\oslash$ denote elementwise multiplication and division; for an absent population, both removal rates are zero. With the vacancy-family lifetime used in the density equation, this gives

$$\mathbf{A}_c = \left[\frac{c_I^{vL}}{\tau_{vL}}, 0, 0, 0 \right]^T. \quad (51)$$

Thus annealing removes cluster number and stored content at the same fractional rate, leaving their ratio unchanged under this process alone. The released vacancies return to the mobile pool through $\mathbf{S}_{vL}$ in Eq. (55), conserving defect content during dissolution.

The lifetime τ_{vL} represents dissolution phenomenologically. As written, the terms n_I^{vL}/τ_{vL} and c_I^{vL}/τ_{vL} act on the entire vacancy-loop family; they do not distinguish sub-critical embryos from stable loops. Restricting dissolution to embryos would require an explicit size-dependent factor applied consistently to both removal terms, or a separate embryo population. An absorption gate alone does not impose this restriction.

Vacancy loops that survive and grow lose vacancies through a second, distinct channel: the thermal emission of single vacancies from a stable loop, governed by the size-dependent vacancy binding energy. Because the reduced model tracks only the family density n_I^k and content c_I^k , the emission rate is evaluated at the mean loop size

$$\bar{m}^k = \frac{c_I^k}{n_I^k}, \quad (52)$$

and, following the detailed-balance form of (9)–(11), the per-defect thermal-emission rate of the vacancy loop is

$$\alpha^{v,k}(\bar{m}^k) = \beta_{v,k} \exp\left(-\frac{E_b^{v,k}(\bar{m}^k)}{k_B T}\right), \quad k = vL, \quad (53)$$

where $E_b^{v,k}(\bar{m}^k)$ is the vacancy binding energy of the mean-size loop, whose size dependence must be specified by the emission model. Here $\beta_{v,k}$ must have units of inverse time and be normalized consistently with the per-defect emission rate $\alpha^{v,k}$. This channel reduces the loop content without removing loops, and hence enters the content ODE (40) through the thermal-emission vector

$$\mathbf{A}_c^{\text{emit}} = [\alpha^{v,vL}(\bar{m}^{vL}) c_I^{vL}, 0, 0, 0]^T, \quad (54)$$

which is subtracted alongside $\mathbf{A}_c$ while leaving the density ODE unchanged.

The vacancies released by both channels are returned to the free vacancy pool, sourcing the mobile species PDE (1) through

$$\mathbf{S}_{vL} = \left[\frac{c_I^{vL}}{\tau_{vL}} + \alpha^{v,vL}(\bar{m}^{vL}) c_I^{vL}, 0, 0, 0 \right]^T, \quad (55)$$

whose single non-zero entry feeds the mobile vacancy line. By construction $\mathbf{S}_{vL}$ is term-by-term identical to the vacancy content removed from the loops by $\mathbf{A}_c + \mathbf{A}_c^{\text{emit}}$, so the exchange is a pure transfer: every vacancy debited from the immobile families is credited to the mobile pool exactly once, and the total point-defect content is conserved. This transfer is independent of the mass-conserving absorption coupling $\dot{\mathbf{G}}^c$, so a surviving loop may simultaneously grow by absorbing the excess vacancy flux (debited from $\mathbf{c}_M$ via $\mathbf{D}$) and shrink by thermal emission (credited to $\mathbf{c}_M$ via $\mathbf{S}_{vL}$) without any double counting. The nucleation content is debited through the second-order mobile reactions, whereas absorption is debited through $\mathbf{D}\mathbf{c}_M$. These are distinct transfers. Opposite-polarity absorption annihilates one mobile defect and one stored defect; it therefore conserves signed defect content, while reducing the total number of defects by recombination.

3 The ZrMicro Code and Parameter Identification

ZrMicro provides the 0D approximation and parameter-identification tools used to calibrate **MoDELib2**. It is delivered as an independent repository. Whereas **MoDELib2** resolves the spatial fields described in Section 2, **ZrMicro** follows spatially uniform defect populations. Its lower computational cost makes it practical to repeat the simulation for many trial parameter sets and compare the predictions with experimental data. The resulting parameters are supplied to **MoDELib2** through `Zr_CD4opt.txt`.

3.1 The ZrMicro model

The 0D model predicts how mobile defects and dislocation-loop populations evolve with irradiation dose and temperature. It represents vacancies, interstitials, di-interstitials and tri-interstitials, together with the number density and stored defect content of prismatic interstitial loops and basal vacancy loops. Each loop type is divided into aligned and non-aligned populations for stress-dependent nucleation. These are populations within one model, not separate codes. The reported calibration uses zero applied stress.

The model integrates 19 ordinary differential equations: four mobile-species concentrations, four loop number densities, four loop contents, the network dislocation density, and six accumulators used to check defect conservation. In compact form,

$$\frac{d\mathbf{y}}{dt} = \mathbf{f}(\mathbf{y}; \mathbf{p}, T), \quad \mathbf{y}(0) = \mathbf{y}_0(\mathbf{p}, T), \quad \text{dpa} = Gt, \quad (56)$$

where $\mathbf{p}$ is the material-parameter vector, T is temperature and G is the dose rate. A stiff time integrator follows both the rapid adjustment of mobile defects and the slower evolution of the loops. The compiled solver uses CVODE/BDF; a Python implementation using LSODA is also available.

Unlike the quasi-steady mobile-field calculation in `MoDELlib2`, the 0D model advances the mobile concentrations in time and contains no spatial transport term. It therefore cannot resolve a grain-boundary depletion layer. Its outputs describe a uniform material volume and provide a reference for the interior of a spatial calculation, where boundary effects are small.

The notation also differs: the 0D mobile variables C_v, C_i, C_{2i}, C_{3i} count clusters per host atom. Their defect contents are consequently $C_v, C_i, 2C_{2i}, 3C_{3i}$, respectively. The fields c_M^m in Section 2.2 already count defects per host atom. This conversion must be included when comparing the two codes.

The kinetics include cascade production, mobile-defect reactions, absorption at loops and the dislocation network, loop nucleation, dissolution, and coalescence. Temperature-dependent jump frequencies determine the reaction rates and are related to diffusivities by Eq. (8). Network sinks are represented by $C_s^v = (a^2/z_c)\rho_N$ and $C_s^i = (a^2/z_c)Z_N\rho_N$. Loop absorption is accounted for separately, so it is not included again in these network sink concentrations. The mobile losses and the corresponding loop-content changes are tracked together in the defect balance.

In the preliminary calibration, the effect of anisotropy on capture is represented by fitted bias factors,

$$Z_i^a = 1 + \delta_i(T), \quad Z_v^a = 1 - \delta_v, \quad Z_i^c = 1 - \delta_i(T), \quad Z_v^c = 1 + \delta_v. \quad (57)$$

The interstitial parameter is interpolated between fitted temperature anchors; the vacancy parameter is held constant. These effective capture factors accompany isotropic diffusion in the present spatial calculations. The calibration will be updated to include anisotropic diffusion in the next deliverable.

Coalescence limits the loop population through two mechanisms. Like-loop coalescence reduces loop number while preserving stored content. Loop-network coalescence removes both and contributes dislocation line length to the network. For a family with number concentration n , volume density $N = n/\Omega$, and mean radius r , define the spacings $d_{LL} = N^{-1/3}$ and $d_N = \rho_N^{-1/2}$, and the capped radii $r_{LL} = \min(r, d_{LL})$, $r_{LN} = \min(r, d_N)$. The probabilities and rates referenced in Section 2.4 are

$$\phi_{LL} = 1 - \exp(-\kappa_{LL} \frac{4}{3} \pi r_{LL}^3 N), \quad \phi_{LN} = 1 - \exp(-\kappa_{LN} \pi r_{LN}^2 \rho_N), \quad (58)$$

$$\nu_{LL} = c_{LL} v_{\text{abs}} N^{1/3}, \quad \nu_{LN} = c_{LN} v_{\text{abs}} \sqrt{\rho_N}. \quad (59)$$

Here κ_{LL}, κ_{LN} control the overlap probabilities, c_{LL}, c_{LN} scale the rates, and v_{abs} is the climb speed associated with absorbed like defects. Using absorbed flux rather than net growth allows coalescence to continue when opposing fluxes balance. The corresponding losses are $\Delta N = (\nu_{LL} \phi_{LL} + \nu_{LN} \phi_{LN})n$ and $\Delta C = \nu_{LN} \phi_{LN} c$. The network density evolves as loops join it and as recovery relaxes it toward its initial value.

For each loop family, the mean radius is reconstructed from its content-to-number ratio, $r = \sqrt{\Omega c / (\pi b n)}$, using the appropriate Burgers-vector magnitude b . The quantities compared with experiment are the total number density and number-weighted mean diameter of each loop type, combining its aligned and non-aligned populations. Full dose histories, conservation checks, and comparisons with measurements are reported in Section 4.

3.2 Parameter identification and transfer to MoDELlib2

Parameter identification adjusts the uncertain kinetic parameters so that **ZrMicro**’s predicted loop densities and diameters agree with measurements across temperatures and doses. The input workbook **Zr_input_parameters.xlsx** contains the experimental targets, initial parameter values, and allowed search ranges. Its **Params** worksheet specifies which parameters are fitted and which remain fixed.

The experimental targets are separated by loop type. **Targets_A** contains 47 usable prismatic-loop records over nine temperatures from 473 to 673 K, with 45 density and 35 diameter measurements. Diameter measurements at $T \geq 650$ K that correspond to vacancy-character prismatic loops are excluded from the interstitial-loop fit. **Targets_C** contains five basal vacancy-loop records at 573 and 583 K, each reporting density and diameter. A single simulation at each temperature supplies predictions at all measured doses for both loop types.

The fitting criterion compares logarithmic loop densities and relative diameter errors. Errors are averaged within each loop-type/temperature group before combining the groups, preventing densely sampled conditions from dominating the fit. The high-temperature groups receive additional weight. A further penalty discourages excessive overshoot followed by collapse of the predicted loop histories. The optimization problem is

$$\mathbf{p}^* = \arg \min_{\mathbf{p}_{\min} \leq \mathbf{p} \leq \mathbf{p}_{\max}} J(\mathbf{p}), \quad J = J_{\text{data}} + w_{\text{os}} J_{\text{os}}. \quad (60)$$

Parameters spanning wide ranges are searched logarithmically, and the search coordinates are scaled to avoid differences in units dominating the numerical optimization.

The software offers differential evolution for global exploration, a surrogate-assisted search that learns an inexpensive approximation of the fitting criterion, and local Nelder–Mead refinement. The surrogate is updated whenever a new full simulation is evaluated; it reduces the number of expensive model calls. The first-round results reported here use the surrogate search followed by local refinement. Each candidate is evaluated across all target temperatures, so one parameter set must explain the combined database.

The first round fitted 23 parameters and reached an objective of approximately $J = 0.61$. Table 1 records the values and bounds. Seven parameters reached a search bound and were subsequently held fixed, leaving 16 active parameters for further fitting. Reaching a bound is a reason to review the parameter range or seek independent constraints; an interior optimum alone does not establish that a parameter is uniquely identifiable.

The preliminary fit reproduces the main temperature dependence of loop density and the low-to-intermediate-temperature prismatic-loop diameters. Its largest remaining discrepancy is the basal-loop diameter: about 70 nm compared with measurements of 95–150 nm, despite reasonable agreement in basal-loop density. The results section presents these comparisons and the remaining limitations.

The calibrated values are transferred to **Zr_CD4opt.txt**, the material file read by **MoDELlib2**. This step includes the required unit and concentration conversions and the mapping from jump-frequency reaction rates to the diffusivity-based coefficients used by the spatial model. Matching numerical parameter values alone is insufficient: assembled rates and interior simulation results must also be compared. Those checks are reported in Section 4.2. Thus the delivered workflow has two codes: **ZrMicro** identifies the parameters, and **MoDELlib2** uses the calibrated material file for spatial simulations.

4 Results

The results follow the two-code workflow described in Section 3. First, **ZrMicro** is evaluated against the experimental loop data used for parameter identification. Next, **MoDELlib2** uses the calibrated material file **Zr_CD4opt.txt** to simulate spatial defect evolution. Comparisons with the 0D results assess the transfer of the calibration before examining the spatial variations near grain boundaries. The calibration is preliminary and uses isotropic diffusion; anisotropic diffusion will be addressed in the next deliverable.

Table 1: Preliminary calibration: the 23 fitted parameters, search bounds, and fitted values. “log” denotes logarithmic search; a bullet marks a value at a search bound. The seven marked parameters are held fixed in the subsequent 16-parameter search.

Parameter	min	optimal	max	log	rail
δ_i^{300} (delta_DAD_i_300)	0.050	0.1533	0.450		
δ_i^{600} (delta_DAD_i_600)	0.020	0.0641	0.300		
Z_{iL} (Z_iL)	1.010	1.3441	1.500		
c_{LL}^a (c_LL_a)	0.10	120.58	1000	log	
c_{LN}^a (c_LN_a)	0.01	1130.8	10000	log	
ε_{iL} (epsilon_iL)	10^{-3}	2.11×10^{-3}	0.300	log	
n_{iL}^{nuc} (n_iL_nuc)	5.0	300.0	300		•
ε_{2i} (epsilon_2i)	10^{-4}	0.0100	0.010	log	•
ε_{3i} (epsilon_3i)	10^{-4}	0.0100	0.010	log	•
E_i^m (E_m_i)	0.20	0.7591	2.000		
E_{2i}^m (E_m_2i)	0.20	1.3629	2.000		
E_{2i}^B (E_b_2i)	0.20	0.9570	3.000		
δ_v (delta_DAD)	5×10^{-3}	0.1079	0.400	log	
c_{LL}^c (c_LL_c)	0.01	0.1625	100	log	
c_{LN}^c (c_LN_c)	0.01	2.9583	100	log	
ε_{vL} (epsilon_vL)	5×10^{-3}	5.00×10^{-3}	0.100	log	•
n_{vL}^{nuc} (n_vL_nuc)	5.0	400.0	400		•
E_a^{vL} (E_a_vL)	0.20	0.4701	0.800		
τ_{vL}^0 (tau_vL0)	0.01	1.0514	5.000	log	
Q (Q)	0.10	0.2879	1.000		
Z_N (Z_N)	1.010	1.0100	1.300		•
$c_{\rho N}$ (c_rhoN)	10^{-3}	1.00×10^{-3}	1.000	log	•
$k_{\rho N}^{\text{rec}}$ (k_rhoN_rec)	10^{-9}	3.48×10^{-5}	10^{-3}	log	

4.1 ZrMicro: 0D calibration results

The reduced (spatially averaged, 0D) rate-theory model integrates the coupled cluster-dynamics equations for the mobile point defects and the four loop families—interstitial “a” loops and vacancy “c” loops, each resolved into a randomly oriented population and a diffusional-anisotropy-difference (DAD) aligned population—against an evolving network dislocation density ρ_N . This section reports the converged trajectories obtained at the optimal parameter set of Table 1 and confronts them, family by family, with the assembled Zr loop, growth, and hardening database. Unless noted, all quantities are plotted against dose in displacements per atom (dpa) on a logarithmic scale spanning 10^{-6} to 10^2 dpa, so that the nucleation transient, the overshoot, and the saturation plateau are all visible on a single axis. The recurring qualitative pattern—an early power-law build-up, a mild overshoot near 10^{-2} dpa, and a temperature-controlled saturation plateau—is the signature of the balance between cascade production, mutual recombination, and biased absorption at the growing sink population, and it reproduces the canonical behavior seen in the irradiated-Zr literature.

The point-defect kinetics come first. Figure 1 shows the evolution of the mobile and trapped defect concentrations. The free vacancy concentration C_v rises first and saturates near $5 \times 10^{17} \text{ cm}^{-3}$, while the free interstitial concentration C_i remains two to three orders of magnitude lower because the faster-migrating interstitials are absorbed almost as quickly as they are produced; the di- and tri-interstitial clusters (C_{2i}, C_{3i}) stay flat and subdominant, confirming that small mobile clusters are not a significant storage channel. The defect content bound in loops behaves oppositely: the interstitials stored in *a*-loops and the vacancies stored in *c*-loops climb steeply from $\sim 10^{14}$ to $\sim 10^{20} \text{ cm}^{-3}$ across 10^{-3} –1 dpa and then plateau, so that by saturation essentially all of the retained damage resides in the loop microstructure rather than in the matrix. This separation of the free and trapped populations is the physical basis of the partitioning used in the growth and hardening calculations.

The total stored-defect curve of Figure 2 aggregates these channels and makes the buildup, overshoot, and saturation sequence explicit, while Figure 3 resolves how the interstitial budget is apportioned be-

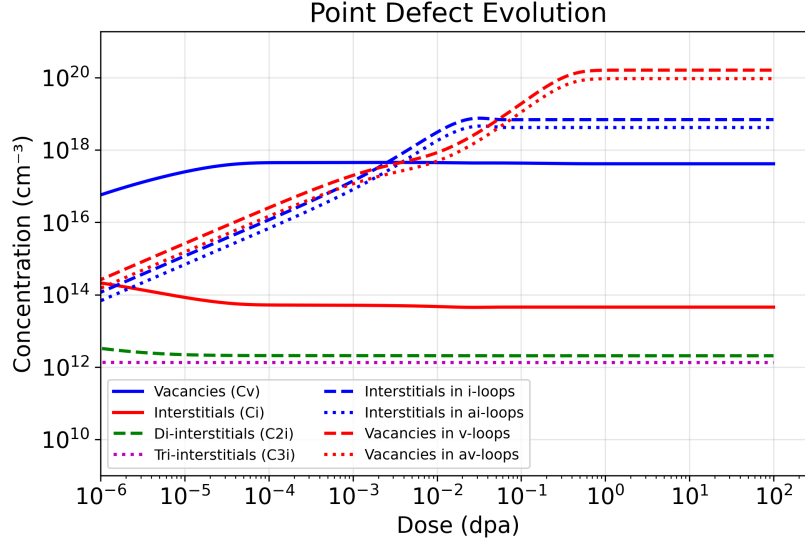


Figure 1: Point-defect evolution. Free vacancies (C_v) saturate near $5 \times 10^{17} \text{ cm}^{-3}$ while free interstitials (C_i) sit well below them; the di-/tri-interstitial channels remain negligible. The interstitial content of a -loops and the vacancy content of c -loops grow by several orders of magnitude before saturating, showing that the retained damage is overwhelmingly stored in the loop population.

tween the randomly oriented and DAD-aligned loop sets. The aligned fractions stabilize at the values set by the orientation bias, which is precisely the partition that produces a net directional strain. Figure 4 plots the running difference between vacancies and interstitials delivered to the sinks; the small but nonzero, sign-definite imbalance is the engine of irradiation growth, and its saturation mirrors the saturation of the loop populations.

The remaining transport diagnostics confirm that the integration is physically self-consistent. Figure 5 tracks the absolute vacancy and interstitial fluxes to the sinks as they rise during the transient and lock together once the sink strength saturates.

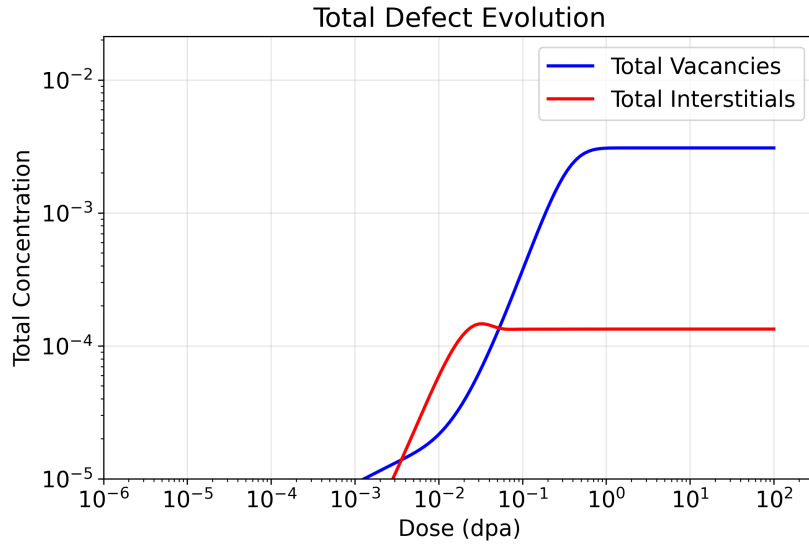


Figure 2: Total stored-defect evolution versus dose, showing the early build-up, the mild overshoot near 10^{-2} dpa, and the saturation plateau that is inherited by all downstream microstructural quantities.

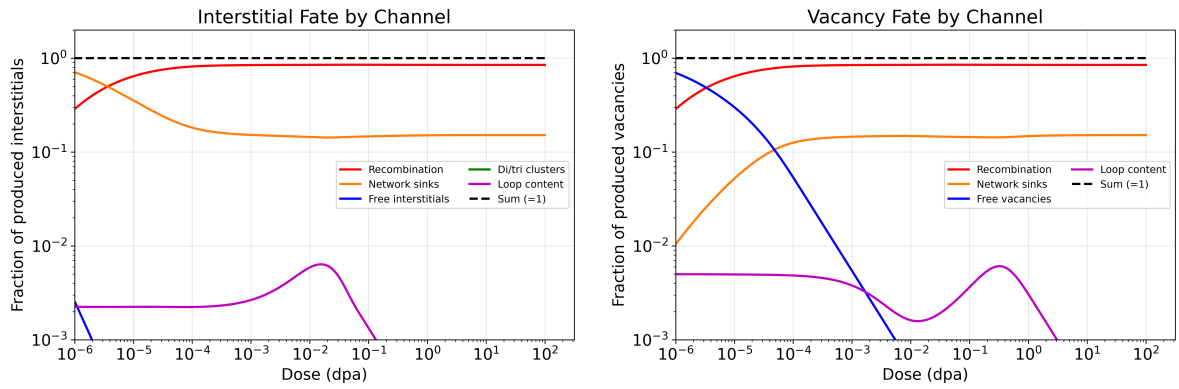


Figure 3: Partition of the interstitial (left) and vacancy (right) budgets between the randomly oriented and DAD-aligned loop populations. The aligned fractions settle to the values fixed by the orientation bias, supplying the directional sink strength that drives growth.

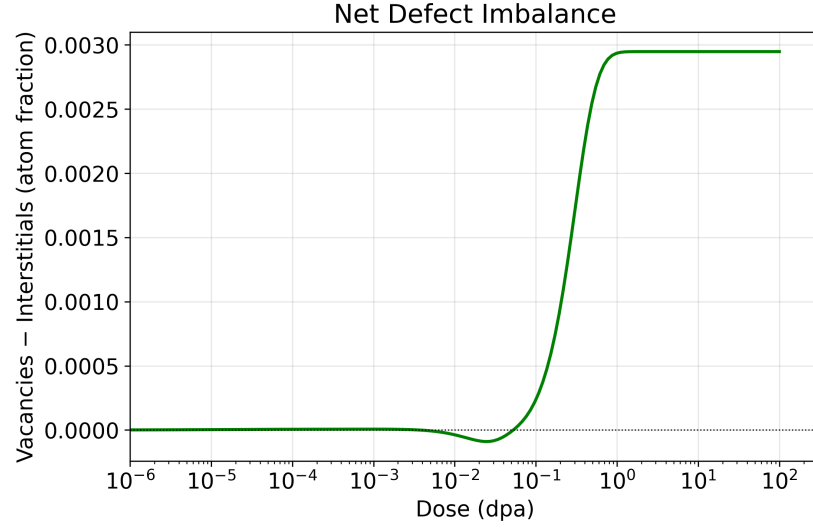


Figure 4: Net vacancy-interstitial imbalance arriving at the sinks. The sign-definite, saturating imbalance is the microscopic driver of the macroscopic growth strain.

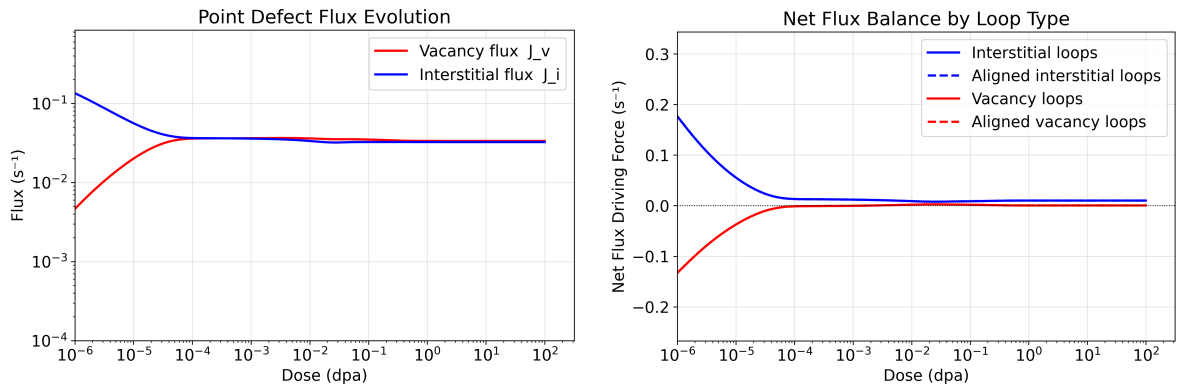


Figure 5: Defect fluxes to the sinks. Left: the absolute vacancy and interstitial fluxes rise during the transient and saturate together. Right: the small, sign-definite net flux produced by the orientation bias, which controls the growth rate.

The loop densities and sizes are the quantities most directly constrained by the experimental database. Figure 6 shows that both loop families nucleate between 10^{-5} and 10^{-2} dpa, pass through a modest overshoot near 2×10^{-2} dpa, and saturate—the interstitial (a) loops near $4 \times 10^{21} \text{ m}^{-3}$ and the vacancy (c) loops near 10^{21} m^{-3} , with the DAD-aligned subsets lying a factor below their parent populations. These plateaux fall within the scatter of the measured a -loop densities of Northwood, Jostsons, Kobylansky, and co-workers, which cluster around $1\text{--}3 \times 10^{22} \text{ m}^{-3}$; the model sits at the lower edge of this band, reflecting the conservative coalescence rate. The companion size plot, Figure 7, shows the a -loop diameter saturating near 9–10 nm, in good agreement with the bulk of the low- and mid-temperature data, while the c -loop (vacancy) diameter climbs after $\sim 10^{-1}$ dpa and saturates near 70 nm. The measured c -loop sizes of Harte (573 K) and Tournadre (583 K) lie at 95–150 nm, so the model under-predicts the c -loop diameter by roughly a factor of 1.5–2; this remains the principal quantitative discrepancy and motivates the targeted c -loop refit. Figure 8 breaks the saturated population down by family and orientation and confirms that the interstitial loops dominate the number density while the vacancy loops dominate the stored volume, consistent with the anisotropic-bias picture.

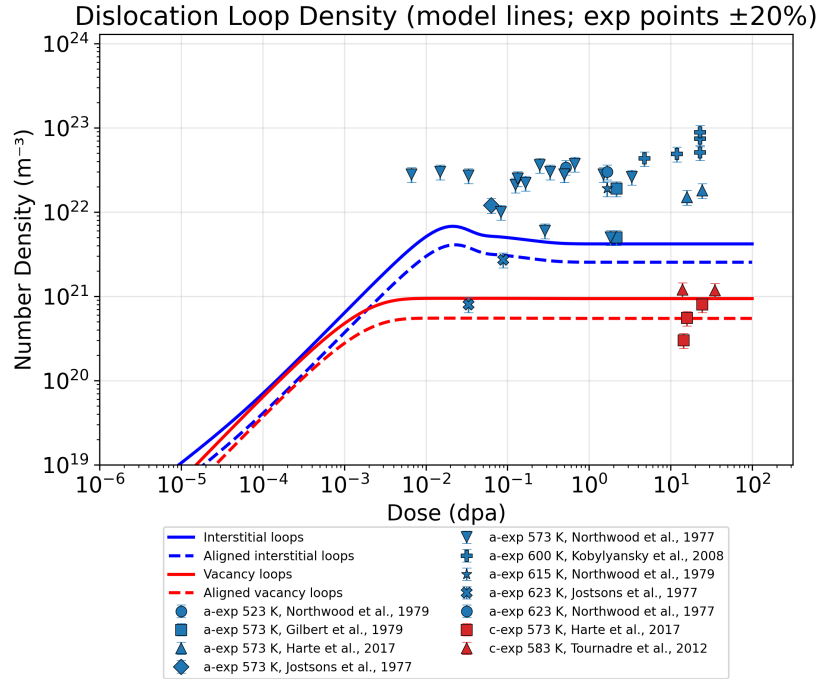


Figure 6: Dislocation-loop number density versus dose (model lines; experimental points with $\pm 20\%$ bars). Interstitial and vacancy loops, each with their DAD-aligned subset, nucleate, overshoot near 2×10^{-2} dpa, and saturate. The a -loop plateau sits at the lower edge of the measured $1\text{--}3 \times 10^{22} \text{ m}^{-3}$ band.

The temperature-resolved comparisons make the agreement quantitative. Figure 9 overlays the model curves, color-coded by temperature, on the a -loop density and diameter measurements between 473 and 673 K. The predicted saturation density decreases monotonically with increasing temperature—from the cold 473 K curve down to the hot 668 K curve—reproducing the well-known thermal suppression of loop number density through enhanced vacancy mobility and recombination, and the model passes through the bulk of the color-matched data. The a -loop diameter saturates near 8–11 nm at all temperatures, matching the low- and mid-temperature points; the large 30–62 nm diameters reported at 668–673 K are the vacancy-character “ a ” loops that are deliberately excluded from the interstitial fit and therefore lie above the model line. Figure 10 presents the corresponding c -loop comparison: the predicted c -loop density saturates near $1.3 \times 10^{21} \text{ m}^{-3}$, matching Tournadre (583 K) and bracketing the Harte (573 K) points, whereas the c -loop diameter again falls short of the measured values, consistent with the dose-resolved view above.

The evolving network dislocation density and the geometric spacing of the sinks complete the microstructural description. Figure 11 shows ρ_N relaxing from its grown-in seed and saturating,

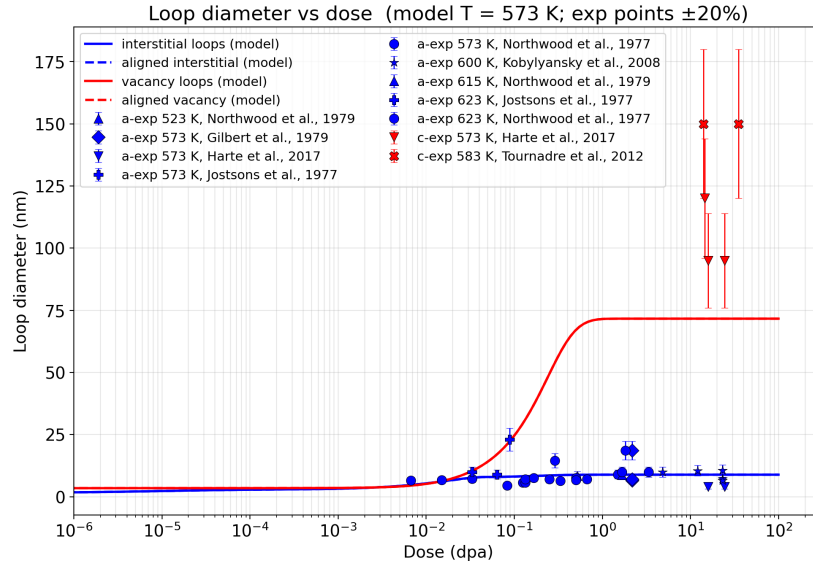


Figure 7: Loop diameter versus dose. The *a*-loop (interstitial) diameter saturates near 9–10 nm in agreement with the low/mid-*T* data; the *c*-loop (vacancy) diameter saturates near 70 nm, below the measured 95–150 nm, the main remaining discrepancy.

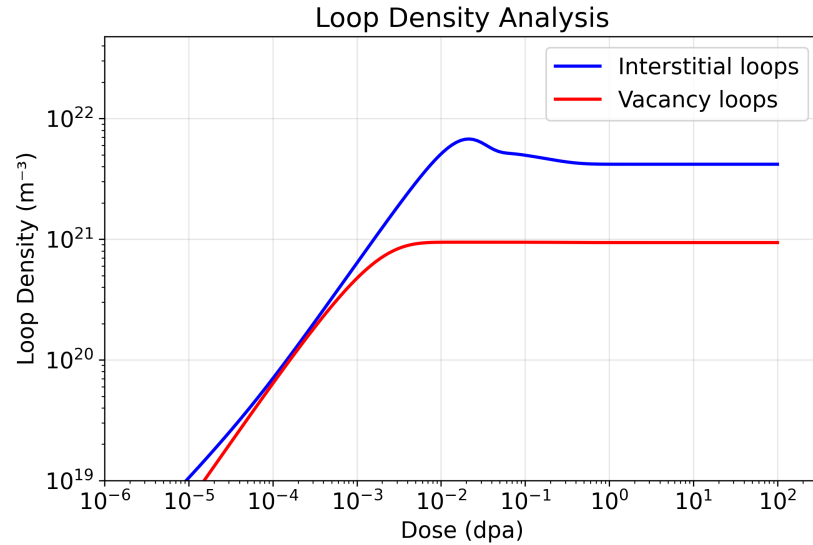


Figure 8: Decomposition of the saturated loop population by family and orientation, showing that interstitial loops control the number density while vacancy loops control the stored defect volume.

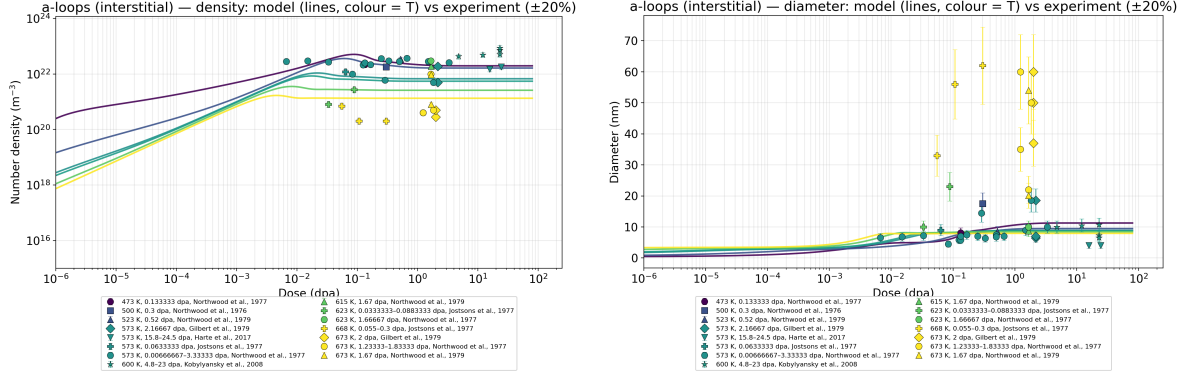


Figure 9: Interstitial (*a*) loops versus the temperature-resolved database. Left: number density; the saturation level falls with increasing temperature, tracking the measured thermal suppression. Right: diameter, saturating near 8–11 nm; the large high-*T* points are excluded vacancy-character loops.

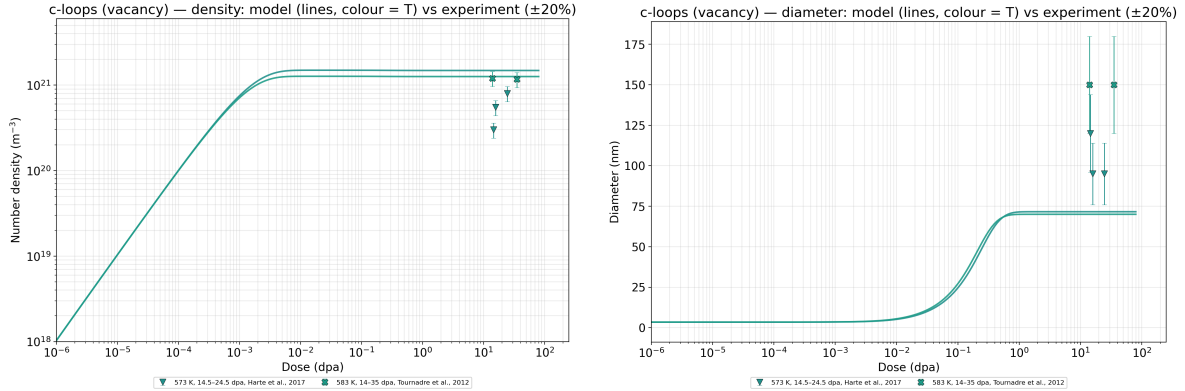


Figure 10: Vacancy (*c*) loops versus experiment. Left: the density saturates near $1.3 \times 10^{21} \text{ m}^{-3}$, matching Tournadre and bracketing Harte. Right: the diameter saturates below the measured 95–150 nm, the main residual discrepancy.

having absorbed loops that unfault and join the network while coalescence limits further growth; this saturating network is what closes the sink-strength balance at high doses. Figure 12 shows the mean loop-to-network and loop-to-loop spacings, both of which decrease as the loop density builds and then level off at the inter-obstacle distance that sets the hardening and the coalescence rate; their saturation is the geometric counterpart of the density plateau in Figure 6.

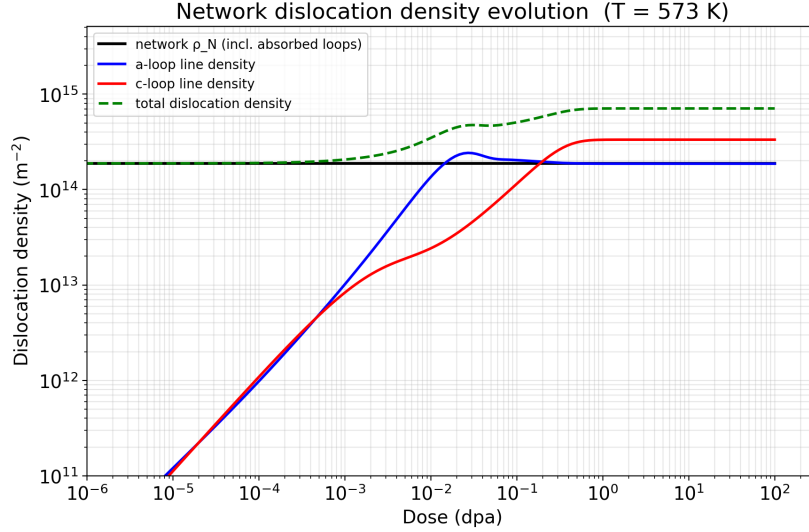


Figure 11: Evolution of the network dislocation density ρ_N , which relaxes from its grown-in seed and saturates as unfaulted loops feed the network and coalescence limits further growth.

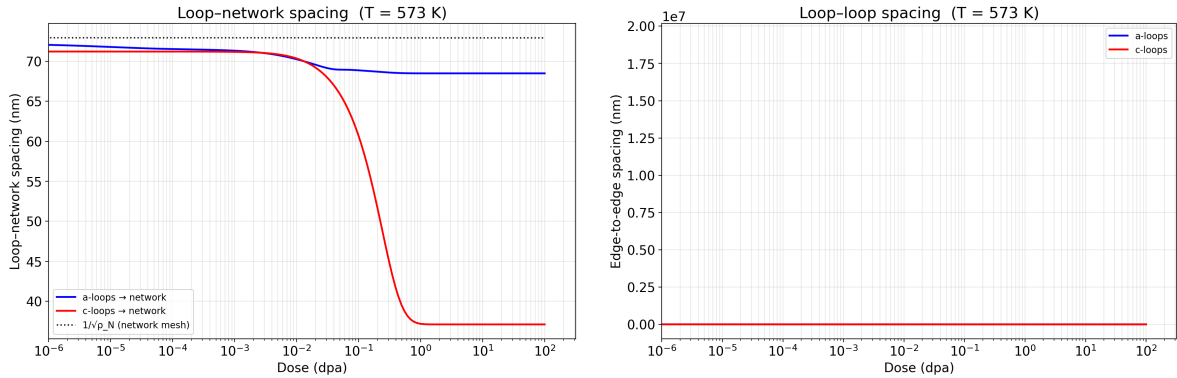


Figure 12: Mean loop-to-network (left) and loop-to-loop (right) spacings. Both contract as the loop density builds and saturate at the inter-obstacle distance that governs hardening and coalescence.

Finally, the conservation diagnostics verify that these results are numerically trustworthy. Figure 13 decomposes the interstitial and vacancy budgets into their production, recombination, and absorption channels, and shows that production is balanced by the sum of the sink channels at every dose. Figure 14 reports the running relative error in the global defect balance, which remains at the round-off level throughout the integration, confirming that the build-up, overshoot, and saturation features above are physical and not numerical artifacts.

4.2 MoDELib2: spatial simulation results

MoDELib2 solves the spatial model of Section 2 using `Zr_CD4opt.txt`. Table 2 compares selected assembled reaction coefficients with those of `ZrMicro`; the following comparison examines the resulting interior concentrations and loop populations.

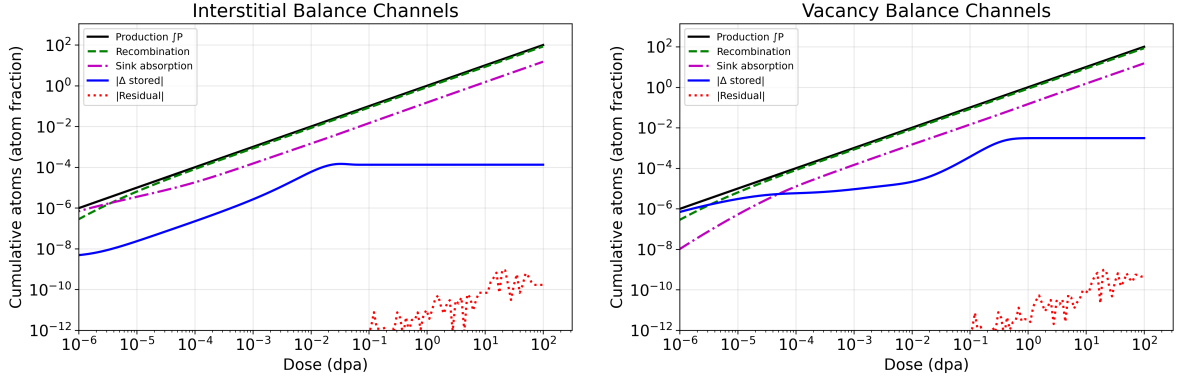


Figure 13: Interstitial (left) and vacancy (right) conservation channels. Production is balanced by recombination plus absorption at the sinks at every dose, confirming closure of the defect budget.

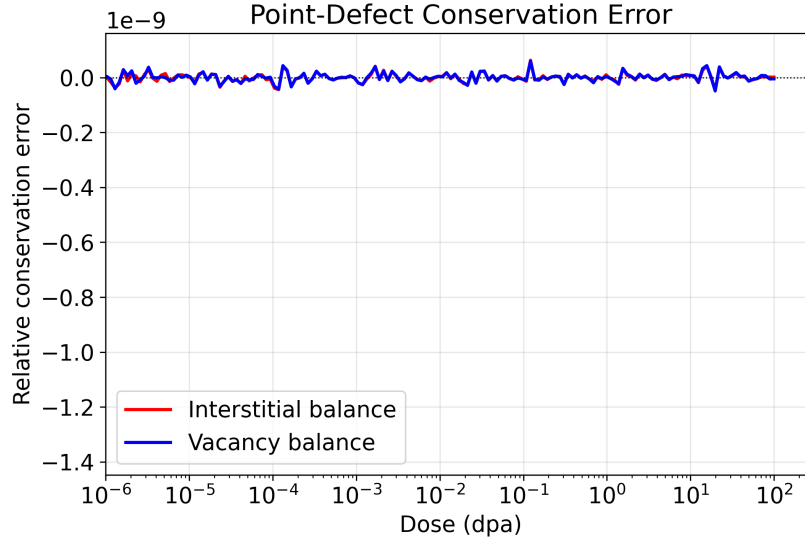


Figure 14: Running relative error in the global defect balance, which remains at the round-off level over the full dose range, establishing that the reported transients and plateau are physical rather than numerical.

Table 2: Assembled second-order interaction coefficients of MoDELlib2, compared with the ZrMicro expressions at 573 K.

entry	MoDELlib2 [s ⁻¹]	ZrMicro [s ⁻¹]	difference
(v, i)	2.526876×10^7	$\frac{1}{2}(\omega_i + \omega_v) = 2.52560 \times 10^7$	0.05 %
(i, i)	-2.021161×10^8	$-4\omega_i = -2.020216 \times 10^8$	0.05 %
$(v, 2i)$	6.940722×10^3	$\omega_{2i} + \omega_v = 6.93473 \times 10^3$	0.09 %

The initial verification results use a cubic domain of 1 μm side at 573 K and $G = 10^{-7}$ dpa/s, with zero applied stress, run to 30 dpa with output every 1 dpa. Every outer face carries the Dirichlet condition, so the entire surface is grain boundary.

Verification against **ZrMicro** comes first. Table 3 lists the ratio of interior 3-D values to the corresponding **ZrMicro** values. Interior nodes are identified from the mobile vacancy field, which saturates away from the absorbing boundary.

Table 3: Ratio of MoDELlib2 interior values to **ZrMicro**, as a function of dose.

dose [dpa]	C_v	C_i	C_{2i}	$N_{\langle c \rangle}$	$C_{\langle c \rangle}$	$N_{\langle a \rangle}$
1	1.073	1.145	1.012	0.999	0.617	0.426
5	0.988	1.124	0.446	0.999	0.647	0.547
10	1.000	1.143	0.421	0.999	0.646	0.540
20	1.008	1.155	0.413	0.999	0.643	0.534
30	1.009	1.159	0.413	0.999	0.642	0.533

The mobile vacancy concentration and the $\langle c \rangle$ loop density agree to better than one percent at 30 dpa. More importantly, every ratio is approximately stable after the initial transient. This indicates that the relative discrepancies stabilize, although constant ratios alone do not distinguish structural differences from systematic errors in rate coefficients or units.

The remaining differences show that transfer of the calibration is not yet quantitatively complete. At 30 dpa, the interstitial concentration is about 16% higher in MoDELlib2, the di-interstitial concentration is about 41% of the **ZrMicro** value, the basal-loop content is about 64%, and the prismatic-loop density is about 53%. The missing homogeneous-clustering contribution to loop nucleation in the reported spatial runs could contribute to the prismatic density deficit, but its magnitude has not been established. The cluster-absorption weighting also requires further checking. Agreement of the selected coefficients in Table 2 does not by itself validate every reaction channel or the full coupled solution.

Turning to the spatial structure the reduced model cannot carry, Fig. 15 shows the four mobile species on a mid-plane cut at four doses. The structure is the classical denuded zone: a plateau in the interior, where production balances absorption at the network and at the loops, falling to the thermal equilibrium concentration at the boundary over a depletion length

$$\ell = \frac{1}{\sqrt{k^2}} \approx 73 \text{ nm} \quad \text{for} \quad k^2 \simeq \rho_N = 1.88 \times 10^{14} \text{ m}^{-2}, \quad (61)$$

which is small compared with the 1 μm domain. The interior is therefore approximately fourteen depletion lengths across, which is what makes the comparison with the well-mixed 0-D model meaningful.

The depleted layer visibly thickens with the dose. This is not a change in the diffusivity but in the sink strength: as loops accumulate, k^2 rises, the interior concentration falls, and the balance between boundary absorption and bulk absorption shifts.

The immobile families follow the same pattern. Figures 16 and 17 show the four immobile families. Their spatial structure is the inverse of the mobile fields, and the reason is instructive. Cascade nucleation is spatially uniform, being proportional only to the dose rate. Loop removal, however, occurs by coalescence, whose rate is set by the absorbed mobile flux. Near the boundary that flux vanishes, so loops are created at the full rate and never removed: the density rises steeply while the size of each loop falls.

The two loop types then behave in opposite senses, and the distinction matters for the growth strain. For the $\langle a \rangle$ families the density rises by more than three orders of magnitude toward the boundary while the mean number of defects per loop falls only by a factor of about six, so the density excess dominates and the stored content is enhanced at the boundary. For the $\langle c \rangle$ family the density is set by the balance between cascade nucleation and dissolution, Eq. (35), which involves no mobile flux at all and is therefore almost uniform in space; the size deficit is then unopposed and the stored content is depleted at the boundary. Both behaviors are visible in Fig. 22.

The three prismatic panels in Fig. 16 are identical, as they must be at zero applied stress. This is a sensitive internal consistency check for the spatial calculation: an error in the Burgers vector components produced clearly unequal densities and was detected this way.

Zr3d_ghoniem: mobile defect concentrations on a mid-plane cut

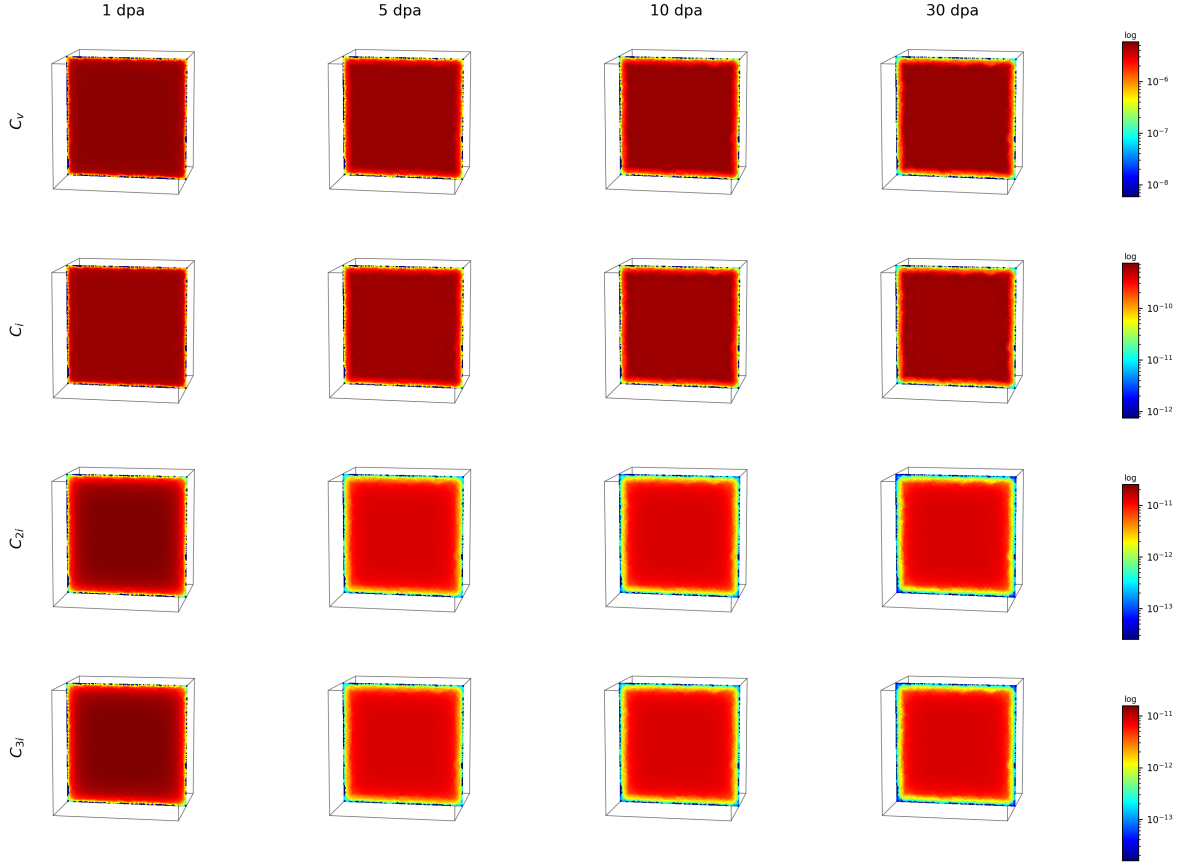


Figure 15: Mobile defect concentrations C_v , C_i , C_{2i} and C_{3i} on an interior mid-plane cut, at 1, 5, 10 and 30 dpa. The color scale is logarithmic over three decades below the maximum, which places the color range across the depletion shell rather than on the interior plateau. The depleted layer thickens with dose as the loop sink strength grows.

Zr3d_ghoniem: loop number density on a mid-plane cut

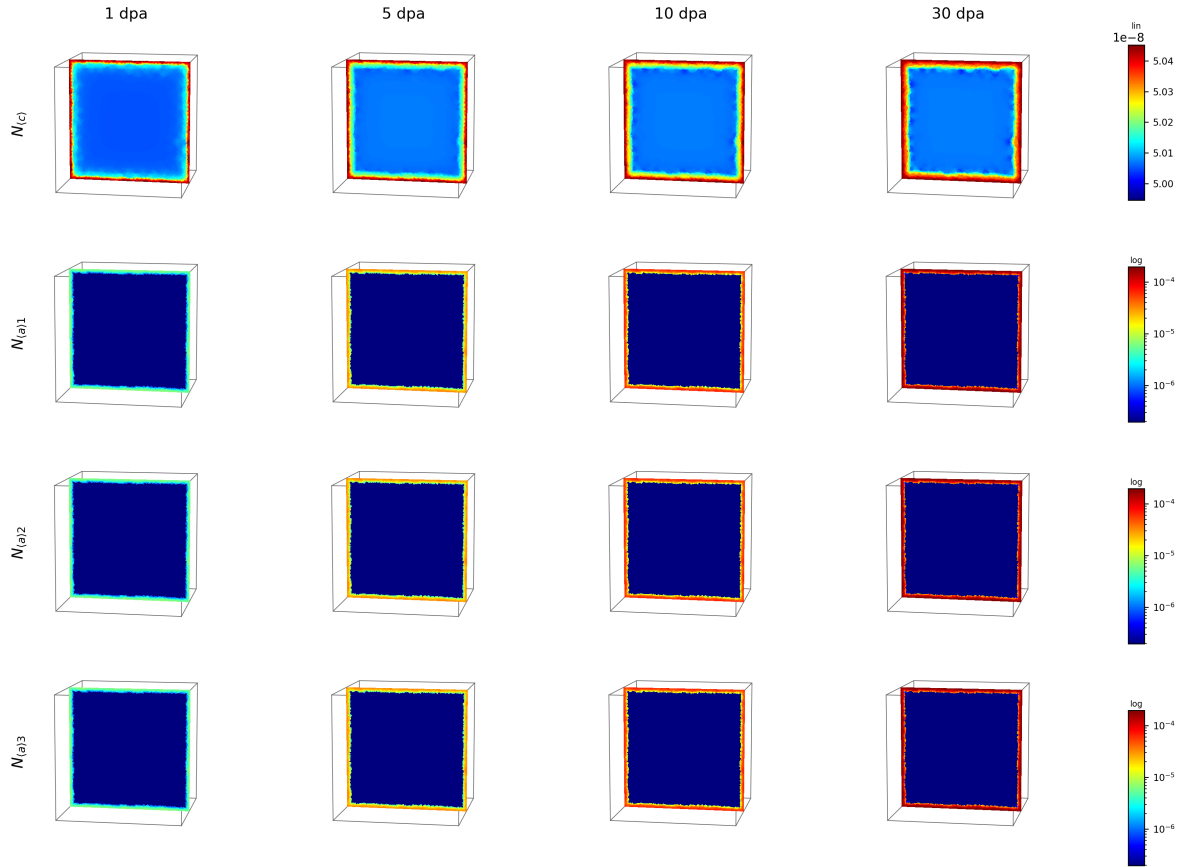


Figure 16: Loop number density for the basal $\langle c \rangle$ family and the three prismatic $\langle a \rangle$ families. At zero applied stress the three $\langle a \rangle$ panels must be identical, and are; this is a check on the crystallography. Note the accumulation adjacent to the grain boundary.

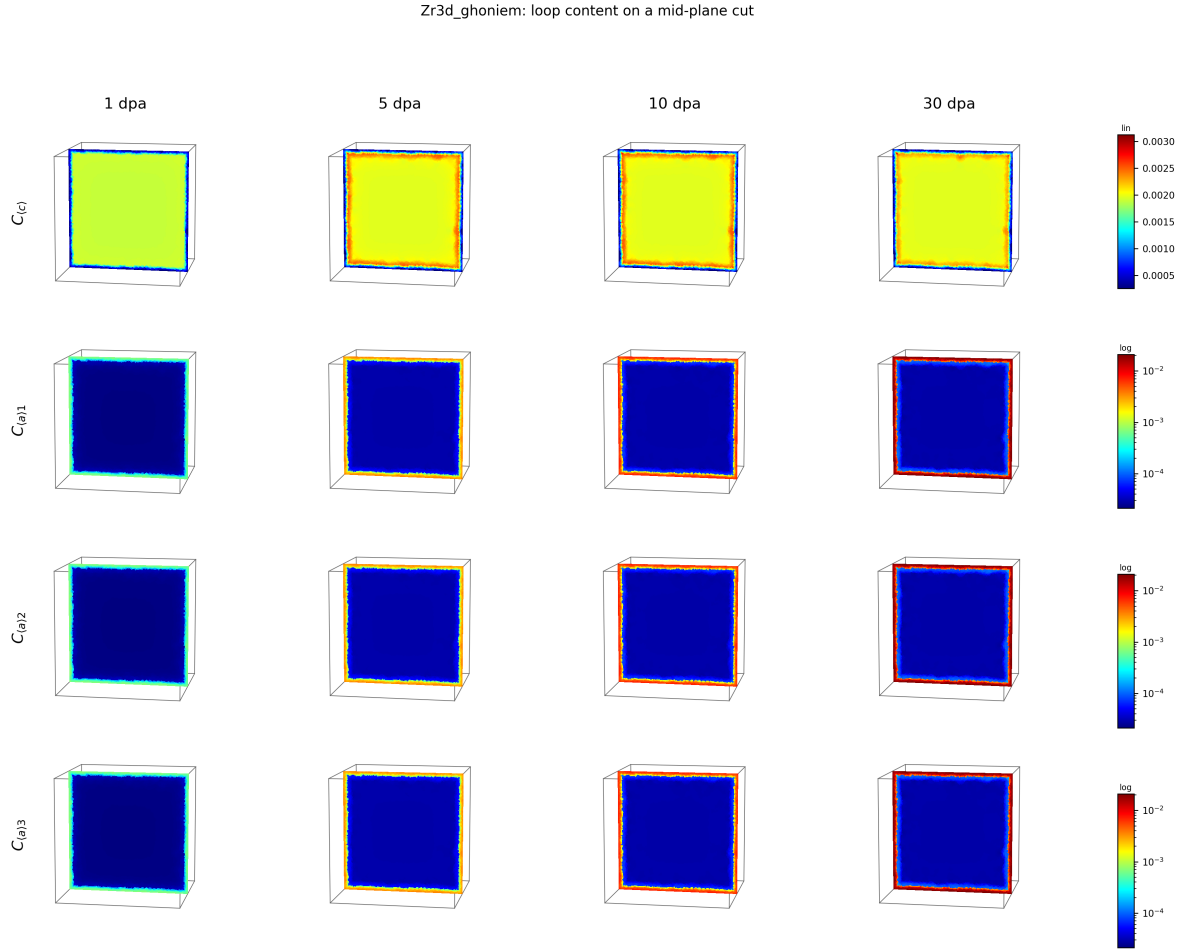


Figure 17: Stored defect content of the four loop families. The $\langle a \rangle$ families are enhanced at the boundary, where the density excess outweighs the size deficit; the $\langle c \rangle$ family is depleted there, its density being almost uniform in space so that the size deficit is unopposed.

The continuum fields can also be drawn as the discrete populations they represent. Figure 18 shows the $\langle c \rangle$ population overlaid on its own content field, with each platelet drawn at the local mean radius and in its correct habit plane. Figure 19 shows one prismatic family. The platelet radii are exaggerated by a factor of 45: the true mean radius is of the order of 4 nm against a 1 μm box, so an honest scale would draw nothing. The relative sizes remain meaningful.

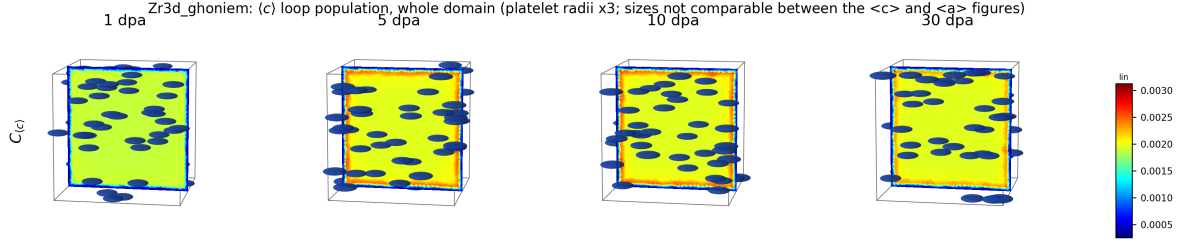


Figure 18: Basal $\langle c \rangle$ vacancy loop population. Platelets lie in the basal plane, consistent with the $\langle c \rangle$ Burgers vector.

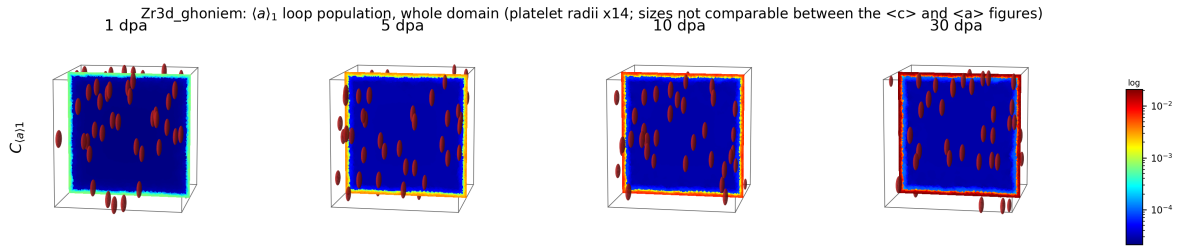


Figure 19: One of the three prismatic $\langle a \rangle$ interstitial loop families. The habit plane differs from Fig. 18, as it must.

The $\langle c \rangle$ population is of particular interest because it controls the c -axis component of irradiation growth. Its number density saturates almost immediately, at the balance $N^* = \dot{N}_{\text{nuc}} \tau_v L$ between cascade nucleation and embryo dissolution, and thereafter the population evolves only through the size of the surviving loops.

The grain boundary is where the spatial model departs furthest from the reduced one, and it is best read as a profile rather than as a field. Figures 20 to 23 present radially averaged profiles against distance from the grain boundary. Because every face carries the Dirichlet condition, the distance is the minimum over all six faces, and each bin averages over the faces and over position within a face.

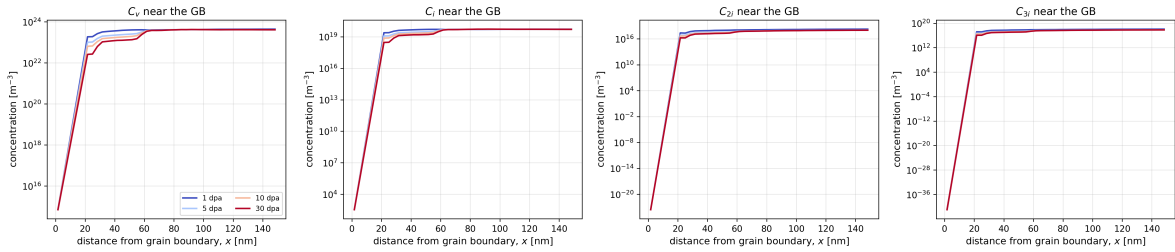


Figure 20: Mobile defect concentrations against distance from the grain boundary. The recovery length is the depletion length $\ell \approx 73 \text{ nm}$; beyond roughly 300 nm the profiles are flat, confirming that the interior is well mixed.

Figures 21 and 23 together give the physical picture of the denuded zone: adjacent to the boundary, there are many small loops, and in the interior, fewer large ones. The total stored content, which is what enters the growth strain, is nevertheless depleted at the boundary (Fig. 22), since the size deficit outweighs the density excess.

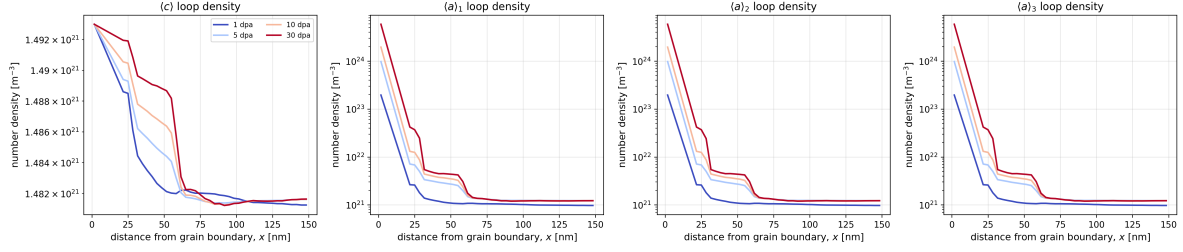


Figure 21: Loop number density against distance from the grain boundary. The $\langle a \rangle$ families rise by more than three orders of magnitude within the first few tens of nanometres.

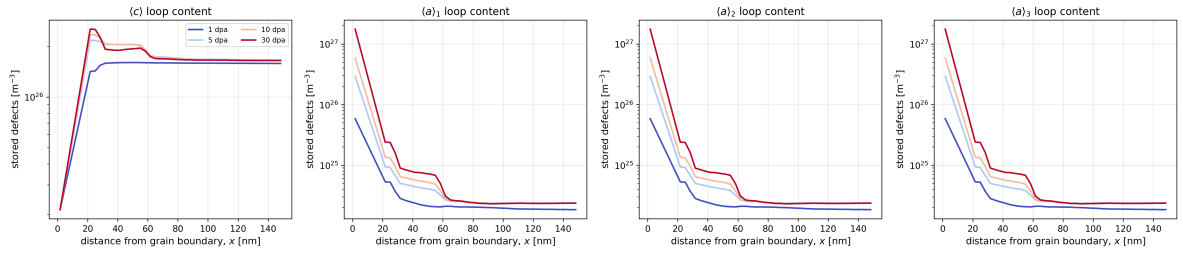


Figure 22: Stored defect content against distance from the grain boundary.

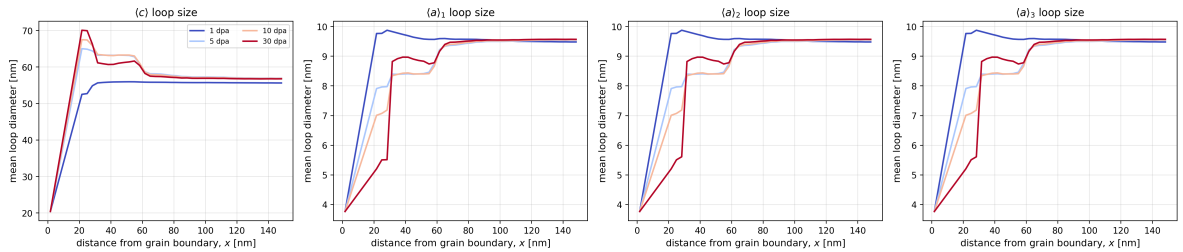


Figure 23: Mean loop diameter against distance from the grain boundary. The diameter is formed from the binned mean content and density, not by averaging nodal diameters: size is a ratio of two fields, and the mean of a ratio is not the ratio of the means.

One caution attaches to this. The $\langle a \rangle$ density adjacent to the boundary grows without bound, linearly with dose. The mechanism is the one described above: uniform cascade nucleation with a removal channel that requires a mobile flux that the boundary condition suppresses. Physically, a grain boundary is also a sink for loops that grow into it, and that channel is absent from the present formulation. The consequence is that the near-boundary loop density should not be used quantitatively, and any grain-size-dependent prediction integrating over the grain will inherit the error. The interior comparison of Table 3 is unaffected, being evaluated well away from the boundary.

The results are checked for mesh dependence on two domains. Figure 24 shows the two finite-element domains used in the mesh-dependence studies. The left panel is the cubic verification domain, meshed with 281,218 tetrahedra graded from 29 nm elements inside a 150 nm boundary layer to 147 nm elements in the interior. The grading follows the structure of the quasi-steady mobile fields: all spatial variation is confined to the point-defect depletion layer adjacent to the absorbing boundary, where the concentrations fall from the interior plateau to the Dirichlet value. Thus, the boundary layer sets the resolution requirement while the flat interior tolerates coarse elements at no cost in accuracy. The right panel is the hexagonal single-crystal domain used for the production runs, 400 nm across and 653 nm tall, meshed with 20,726 tetrahedra. The acceptance criterion is the same for both meshes: interior concentrations and loop observables must be insensitive to further refinement, with several elements across the depletion length. An under-resolved boundary layer misrepresents the grain-boundary sink and contaminates the interior solution through the global defect balance, so convergence is judged on the near-boundary profiles and not only on interior averages.

281,218 tets — 29 nm in the 150 nm layer, 147 nm interior (cut plane facing the viewer)

Hexagonal Zr single crystal, 400 nm wide x 653 nm tall — 20,726 tets (cut normal to y)



Figure 24: Mesh-dependence studies. Left: cubic verification domain, 281,218 tetrahedra, graded from 29 nm in the 150 nm boundary layer to 147 nm in the interior (cut plane facing the viewer). Right: hexagonal Zr single crystal, 400 nm wide by 653 nm tall, 20,726 tetrahedra (cut normal to y).

Figure 25 shows the $\langle a \rangle_3$ loop size distribution at 1 and 26 dpa. Two features carry the physics. First, the distribution is essentially monodisperse at both doses, and the number-weighted mean is stationary at 3.87 nm at 1 dpa against 3.77 nm at 26 dpa, while the bin population grows by more than an order of magnitude, from about 1.3×10^{27} to 3.3×10^{28} loops per m^3 . The prismatic population is therefore nucleation-dominated: cascades inject new loops at the nucleation size faster than the existing loops grow, and additional dose is stored as number density at nearly constant size. Second, the width of the histogram is spatial rather than physical. Each node carries a single mean size per family, so the plotted spread measures the variation of the local mean across the grain; the sharpness of the peak is itself a verification statement, showing that the interior is well mixed, as the reconciliation with the 0-D model requires, while the sparse large-diameter tail is contributed by the few near-boundary nodes whose local density and net flux depart from the interior values.

The $\langle c \rangle$ population of Fig. 26 responds to dose in the opposite way. The distribution is bimodal. The dominant population sits near 55 nm at 1 dpa and coarsens slowly to 58–65 nm by 26 dpa, moving the number-weighted mean from 46.0 to 48.5 nm: basal loops nucleate rarely and at a much larger size than the prismatic loops, and subsequent dose is stored as loop growth rather than as number density. The secondary population near 20 nm is stationary in both position and amplitude between the two doses, which identifies it as spatial in origin: it collects the nodes inside the near-boundary depletion

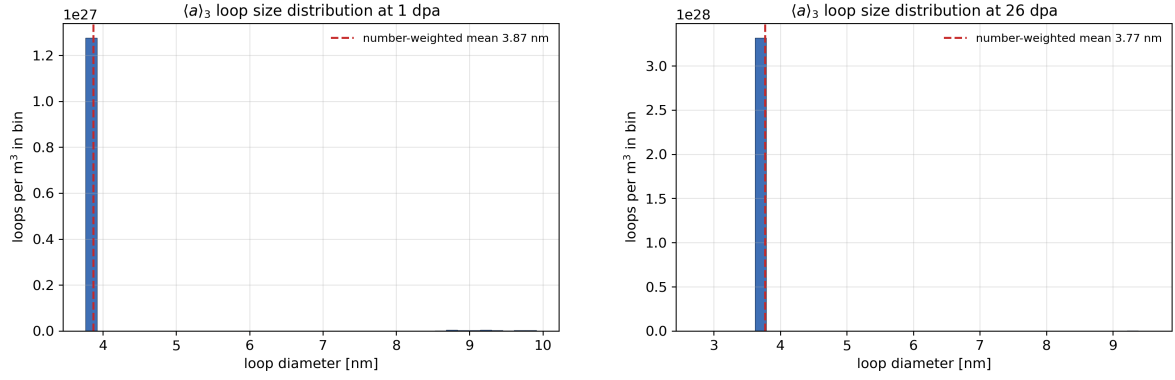


Figure 25: $\langle a \rangle_3$ loop size distributions at 1 dpa (left) and 26 dpa (right). The number-weighted mean diameter is nearly dose-independent while the number density grows by more than an order of magnitude, the signature of a nucleation-dominated population.

zone, where the grain boundary intercepts the mobile fluxes and the local loops are held close to their nucleation size. Its dose independence is consistent with a denuded-zone width set by the depletion length, which does not evolve with dose. The broadening of the interior peak at 26 dpa reflects the accumulated radial gradient of net absorbed flux between grain center and boundary.

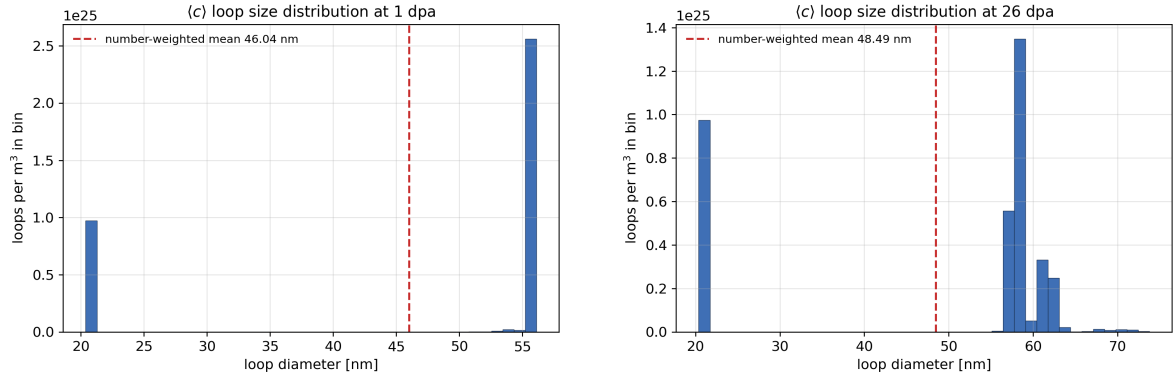


Figure 26: $\langle c \rangle$ loop size distributions at 1 dpa (left) and 26 dpa (right). The interior population coarsens slowly with dose while a stationary near-boundary population remains near 20 nm; the number-weighted mean moves from 46.0 to 48.5 nm.

Figure 27 renders the four loop families in the unstressed hexagonal single crystal at 6 dpa. Platelets are drawn in their habit planes with radii magnified for visibility, by a factor of 7 for the prismatic families and only 1.5 for the basal family; the basal loops are physically an order of magnitude larger. The three prismatic panels constitute the zero-stress crystallography check: with no applied stress, the three $\langle a \rangle$ Burgers-vector families are physically equivalent, and the computed $C_{\langle a \rangle 1}$, $C_{\langle a \rangle 2}$, and $C_{\langle a \rangle 3}$ fields are statistically indistinguishable, each carrying one third of the prismatic population and differing only in habit-plane orientation. Any systematic departure among the three panels would indicate an error in the crystallographic bookkeeping rather than in the physics. The cut planes make the spatial structure quantitative. The prismatic content, plotted on a logarithmic scale spanning 10^{-4} to 10^{-3} , is uniform over the grain interior, with a narrow band of elevated content within the depletion layer at the boundary; the basal content, on a linear scale up to 2×10^{-3} , shows the same flat interior plateau but is denuded at the boundary. The opposite sign of the two near-boundary perturbations reflects the different depletion profiles of the two mobile species within the boundary layer: the net interstitial flux feeding the prismatic loops is locally enhanced where the boundary drains the vacancy field, while the basal vacancy loops lose their growth flux there. Both rims are confined to a few depletion lengths, and the uniformity of all four interior fields is the spatial statement of the reconciliation objective: away from the boundary layer the 3-D solution reduces to the well-mixed

limit described by the 0-D model.

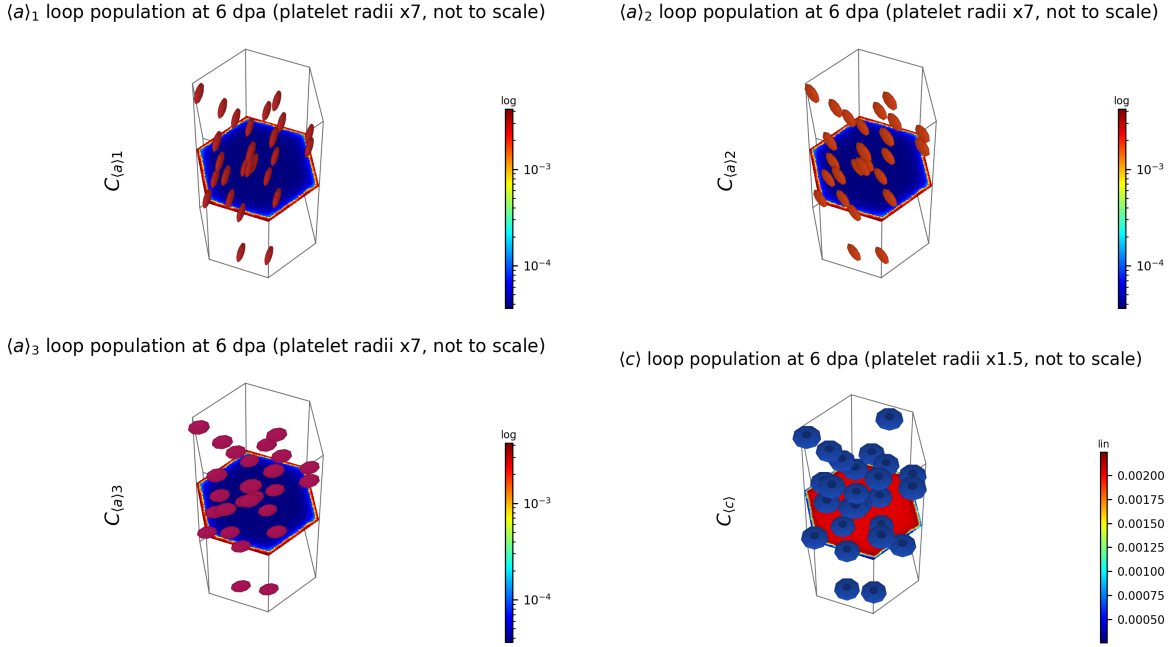


Figure 27: Loop populations in the unstressed hexagonal single crystal at 6 dpa; platelet radii are magnified ($7\times$ prismatic, $1.5\times$ basal) and are not to scale. Top row and bottom left: the three prismatic $\langle a \rangle$ families, whose statistical equivalence at zero stress is the built-in crystallography check. Bottom right: the basal $\langle c \rangle$ family, physically much larger and denuded within the boundary depletion layer.

The comparison links the two delivered codes: **ZrMicro** provides the calibration and uniform reference solution, while **MoDELib2** evaluates that calibration in a finite grain. The selected reaction coefficients agree within 0.1%, and the interior vacancy concentration and basal-loop number density agree within 1% at 30 dpa. Other quantities retain the differences listed in Table 3. The spatial results therefore demonstrate the calibration workflow and grain-boundary effects, while also identifying where further verification is needed.

The following items remain open.

1. **Homogeneous nucleation.** The reported **MoDELib2** runs omit the mobile-clustering contribution to loop nucleation. Adding the source defined in Section 2.4 and repeating the comparison will test its contribution to the prismatic-loop density deficit.
2. **Grain boundaries as loop sinks.** See the caution in Section 4.2.
3. **Di-interstitial consumption.** The dose dependence of the C_{2i} ratio in Table 3 is not understood.
4. **Preliminary diffusion and bias calibration.** The spatial runs use isotropic diffusion with effective capture biases calibrated to the 0D model at 573 K. They do not yet test the tensor-derived DAD factors described in Section 2.3. The next deliverable will update the calibration for anisotropic diffusion.
5. **Software verification.** The comparison identified a binding-energy initialization-order issue in **MoDELib2**. Its correction and verification belong in the delivered **MoDELib2** repository.

5 Conclusions

This deliverable provides two software repositories, **ZrMicro** and **MoDELib2**, the calibrated material file `Zr_CD4opt.txt`, and simulation results for irradiation damage in zirconium. **ZrMicro** supplies the

0D model and parameter-identification tools; MoDELlib2 uses the calibrated inputs to compute spatial defect and loop populations.

The preliminary calibration captures the main trends in loop density and prismatic-loop diameter across the experimental temperature range. Basal-loop diameters remain too small: approximately 70 nm compared with measured values of 95–150 nm. Improving this prediction requires further assessment of the vacancy-loop growth and coalescence model.

The spatial calculations resolve mobile-defect depletion near grain boundaries and differences in loop size and stored content across the grain. Comparison with the 0D model gives good agreement for selected reaction coefficients, interior vacancy concentration, and basal-loop number density. Significant differences remain in prismatic-loop density, basal-loop content, and small interstitial-cluster concentrations. Near-boundary loop densities are not yet quantitative because loop removal at grain boundaries is absent.

The next deliverable will update the calibration to include anisotropic diffusion. Further work will complete the clustering contribution to loop nucleation, include grain-boundary removal of loops, and check the remaining differences between the two codes before using the model for quantitative predictions of irradiation growth, creep, and hardening.

References